Arithmetic Expression Geometry II: Hyperbolic Real Function Theory

Horizontal Operators, Boundary Problems, and Arithmetic Holomorphicity

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Abstract

Paper I of Arithmetic Expression Geometry associates an oriented Riemannian surface, an assignment function, and a canonical arithmetic frame to continuous affine evaluation, and separately constructs a three-dimensional contact state space for ordered additive and multiplicative motion. We develop the analytic structures of these two objects without identifying them. On a regular arithmetic expression space, the metric and orientation determine a complex structure. Writing the frame bracket as $[X_u, X_v] = c_u X_u + c_v X_v$, we compute the formal adjoints and prove the exact frame formula

$$\Delta_g = X_u^2 + X_v^2 - c_v X_u + c_u X_v.$$

The arithmetic Cauchy–Riemann operator then factors the Laplace–Beltrami operator up to a first-order multiple of itself; consequently every arithmetic holomorphic field is harmonic. We also give two functorial surface constructions. A locally biholomorphic map pulls the planar AES $A_\phi(w) = \mathrm{Im}(e^{-i\phi} w)$ back to a regular AES, with harmonic assignment and an explicit curvature law. On $\mathbb{C}^*$, the logarithmic radius supports a complete flat cylindrical AES; its zero circle is the regular analytic prototype for automorphic preimage constructions. On the contact state space we instead normalize the horizontal metric by the preferred lifts. Contact volume gives $D_u^* = -D_u$ and $D_v^* = -D_v - \lambda$, so the variational sub-Laplacian contains a drift omitted by the raw sum of squares. The latter still admits a curvature-twisted factorization, an explicit logarithmic CR field, and finite filtered affine–Appell modules.

For the complete basic hyperbolic AES, the scaled coordinate $X = (\lambda/\mu)x$ identifies the metric with a constant multiple of the standard upper-half-plane metric and turns its arithmetic holomorphic coordinate into $w = X + iy = y(i - (\lambda/\mu)a)$. We prove a C_0 Dirichlet theorem using the Poisson kernel, identify the conformal-boundary Dirichlet-to-Neumann map with the Fourier multiplier $|D|$, and derive its exact energy identity. Rational functions and positive-frequency modes of w give explicit families depending genuinely on the assignment. These results establish a first function theory on the regular hyperbolic model and provide a regular analytic interface to the singular zero geometry of Paper III. Critical points, poles, branched points, and continuation through the resulting singular loci are not treated here; general Green kernels and spectral completeness also remain open.

Keywords: arithmetic expression geometry; hyperbolic plane; Poisson kernel; Dirichlet problem; Dirichlet-to-Neumann map; Cauchy–Riemann equations; contact CR geometry; sub-Laplacian; holomorphic pullback; cylindrical metric

Contents

1 Introduction and principal results

Paper I develops Arithmetic Expression Geometry (AEG) from ordered evaluation histories to affine flow, regular expression surfaces, and contact curvature [4]. The present paper asks the next question: once the geometric data have been fixed, what analysis do they actually support? We answer it first on a general regular arithmetic expression space and then completely enough on the basic hyperbolic model to obtain a boundary kernel, a Dirichlet theorem, and an exact boundary energy operator.

This question has two layers which must not be conflated. A regular AES

$$\mathfrak{E} = (M, g, a; \mu, \lambda)$$

is an oriented Riemannian *surface*. Its metric and orientation already determine rotation by $+\pi/2$, hence the usual complex structure of an oriented Riemannian surface. The assignment a selects a distinguished positively oriented orthonormal frame (X_u, X_v). Complex and elliptic analysis on this surface are therefore intrinsic to the data imported from Paper I.

The contact state space is instead

$$\mathcal{C} = \mathbb{R}^3_{(u,v,a)}, \qquad \alpha = da - \mu\, du - \lambda a\, dv,$$

with preferred horizontal lifts

$$D_u = \partial_u + \mu\partial_a, \qquad D_v = \partial_v + \lambda a \partial_a.$$

When $\mu\lambda \neq 0$, its horizontal distribution is nonintegrable. To do analysis there one must additionally choose a horizontal metric, its compatible complex structure, a positive measure, and operator domains. The resulting object is a three-dimensional contact-CR problem, not another presentation of the two-dimensional surface problem.

Figure 1 records this separation. Both branches retain the arithmetic derivative rules

$$X_u a = \mu, \quad X_v a = \lambda a, \qquad D_u a = \mu, \quad D_v a = \lambda a,$$

but their brackets have different geometric meanings. On a surface the bracket is tangent. On $\mathcal{C}$ one has $[D_u, D_v] = \mu\lambda\partial_a$, a vertical curvature term.

1.1 Principal results

Our first result fixes the real operator theory on a regular AES. If

$$[X_u, X_v] = c_u X_u + c_v X_v,$$

then, with the divergence-of-gradient convention,

$$\Delta_g = X_u^2 + X_v^2 - c_v X_u + c_u X_v. \tag{1}$$

The first-order drift is forced by the Riemannian volume and cannot be discarded. The compatible Cauchy–Riemann operator

$$\overline{\partial}_{\mathrm{AEG}} = \frac{1}{2}(X_u + iX_v)$$

then satisfies an adjoint factorization of $-\Delta_g$. In particular, arithmetic holomorphic fields are genuinely Laplace–Beltrami harmonic. We also show that $\overline{\partial}_{\mathrm{AEG}}$ differs from the ordinary Riemann-surface operator only by a nowhere-zero unitary gauge factor. Thus the surface theory does not

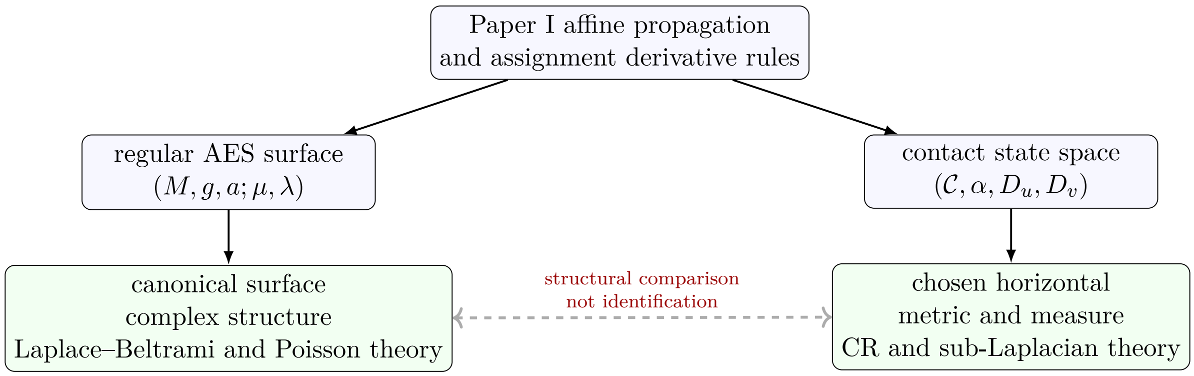

Figure 1: The two analytic objects carried forward from Paper I. The surface branch is elliptic and has an intrinsic complex structure. The contact branch is subelliptic and requires an analytic normalization. Nonintegrability prevents the contact distribution from being identified with a regular AES surface.

manufacture a new class of holomorphic functions: its AEG content is the assignment-adapted frame, the distinguished solution families, and their arithmetic interpretation.

Our second result fixes the corresponding ambiguity on the contact state space. We normalize the horizontal metric by declaring (D_u, D_v) orthonormal and use the positive contact volume. On compactly supported smooth fields,

$$D_u^* = -D_u, \qquad D_v^* = -D_v - \lambda. \tag{2}$$

Consequently the variational sub-Laplacian is

$$\Delta_{\mathcal{C}} = D_u^2 + D_v^2 + \lambda D_v, \qquad -\Delta_{\mathcal{C}} = D_u^* D_u + D_v^* D_v. \tag{3}$$

The legacy expression $D_u^2 + D_v^2$ remains useful, but only as the raw sum of squares occurring in the formal CR factorization. We prove both the raw and adjoint factorizations, relate two natural measures by a unitary gauge transformation, and give a Friedrichs realization of the nonnegative energy operator. Hörmander's bracket criterion supplies hypoellipticity of the differential expression [3].

Between the local operator theory and the global hyperbolic boundary problem, we establish a functorial construction that will be used by the singular theory. The planar data

$$A_\phi(w) = \operatorname{Im}(e^{-i\phi} w), \qquad h_\phi = \frac{|dw|^2}{\mu^2 + \lambda^2 A_\phi(w)^2}$$

form a regular AES with harmonic assignment. Every locally biholomorphic map F pulls this data back to a regular AES; its Gaussian curvature is

$$K = \lambda^2 \frac{\mu^2 - \lambda^2 a^2}{\mu^2 + \lambda^2 a^2}.$$

We also construct the complete flat model on $\mathbb{C}^*$,

$$s = \log |W|, \qquad a = \frac{\mu}{\lambda} \sinh(\lambda s), \qquad g = \left| \frac{dW}{W} \right|^2,$$

with the continuous limiting formula $a = \mu s$ when $\lambda = 0$. Its zero set is the unit circle, and a locally biholomorphic pullback has zero set $W^{-1}(S^1)$. A descent proposition covers unitary automorphy and the possible sign character arising from inversion. The construction is deliberately stated only away from critical points and branch points; the cylindrical construction is in addition restricted away from zeros and poles of W.

Our decisive global theorem concerns the basic hyperbolic AES. For $\mu, \lambda > 0$, put

$$X = \frac{\lambda}{\mu} x, \qquad w = X + iy.$$

Then

$$g_{\mu,\lambda} = \frac{1}{\lambda^2} \frac{dX^2 + dy^2}{y^2}, \qquad \Delta_g = \lambda^2 y^2 (\partial_X^2 + \partial_y^2),$$

and the arithmetic holomorphic equation is precisely $\partial_{\bar{w}} F = 0$. The standard upper-half-plane Poisson kernel therefore transports exactly to the AEG normalization. For every $\varphi \in C_0(\mathbb{R})$ we prove existence and uniqueness in the class $C^2(\mathbb{H}) \cap C(\overline{\mathbb{H}}^{\mathrm{conf}})$ with the prescribed value zero at ∞. For Schwartz data, the conformal-boundary Dirichlet-to-Neumann map is

$$\mathcal{N} = |D_X|,$$

and the hyperbolic Dirichlet energy is exactly its homogeneous $H^{1/2}$ quadratic form. This is the first exact boundary-operator statement of the AEG analytic programme. Its classical analytic ingredients are standard [1, 2]; the result here is their explicit transport, domain statement, and assignment-coordinate interpretation on the Paper I model.

The same model supplies several genuinely assignment-dependent fields because

$$w = y \left(i - \frac{\lambda}{\mu} a \right). \tag{4}$$

Thus every holomorphic $\Phi(w)$ is an explicit field in (a, y) coordinates. We also classify all scalar harmonic fields depending only on a:

$$h(a) = C_0 + C_1 \arctan\left(\frac{\lambda a}{\mu}\right).$$

The nonconstant member is the harmonic measure of a boundary half-line. In the contact branch, logarithmic CR first integrals and finite filtered affine–Appell modules provide a separate collection of assignment-dependent solutions.

1.2 Boundaries of the claim

Several restrictions are essential.

- The line $y = 0$ is the conformal or ideal boundary of the complete hyperbolic metric, not a finite Riemannian boundary component.
- Poisson extensions with nonzero ideal-boundary values are generally not in $L^2(\mathbb{H}, d\mathrm{vol}_g)$. The boundary-energy theorem and the L^2 Friedrichs realization therefore occupy different analytic domains.
- The identity $\Delta_g a = 2\lambda^2 a$ from Paper I is pointwise. The assignment $a = -x/y$ is not an L^2 eigenvector.

- We do not claim that the normalized contact CR structure is invariant under all contactomorphisms, nor that its explicit CR families are complete in a Hilbert or Banach space.
- We do not establish Green kernels, spectral completeness, or continuation on a general regular or singular AES.
- Holomorphic pullbacks are regular AES constructions only where the map is locally biholomorphic. For the cylindrical target, its coordinate W must in addition be finite and nonzero. Critical points, poles, algebraic branch points, and singular zero-graph completion belong to Paper III.

These limitations are not peripheral. They separate proved function theory from the open analytic programme and keep the interfaces to singular geometry and projective complexity acyclic.

1.3 Organization

Section 2 fixes measures, domains, structure coefficients, and energies on a regular AES. Section 3 develops the intrinsic surface Cauchy–Riemann theory and proves holomorphicity implies Laplace–Beltrami harmonicity. Section 4 treats the distinct contact-CR problem. Section 5 proves the planar pullback theorem, constructs the complete cylindrical AES, and states the regular automorphic descent interface. Section 6 identifies all operators on the basic model. Sections 7 and 8 prove the boundary theorems and construct explicit assignment-dependent families. The appendices contain the full frame, contact, and Fourier calculations.

2 Analytic data on a regular arithmetic expression space

Throughout this section

$$\mathfrak{E} = (M, g, a; \mu, \lambda)$$

is a regular arithmetic expression space in the sense of Paper I, with real constants μ, λ. Thus M is an oriented smooth surface, possibly with boundary, g is a smooth Riemannian metric, $a \in C^\infty(M, \mathbb{R})$, $\mu \neq 0$, and

$$|\nabla a|_g^2 = \mu^2 + \lambda^2 a^2. \tag{5}$$

The metric and orientation are part of the imported structure. In particular, they are not reconstructed from an analytic operator introduced later.

2.1 The arithmetic frame and its structure functions

Write

$$q = (\mu^2 + \lambda^2 a^2)^{1/2}.$$

Since $\mu \neq 0$, q never vanishes. Paper I therefore supplies the unique positively oriented orthonormal frame (X_u, X_v) satisfying

$$X_u a = \mu, \qquad X_v a = \lambda a. \tag{6}$$

No coordinate interpretation is implicit here: the frame is generally noncommuting. Define its smooth structure functions $c_u, c_v \in C^\infty(M)$ by

$$[X_u, X_v] = c_u X_u + c_v X_v. \tag{7}$$

Applying this identity to the assignment gives the constraint

$$\mu c_u + \lambda a c_v = \mu\lambda. \tag{8}$$

Equation (8) is only one scalar relation. It does not determine c_u, c_v, the Gaussian curvature, or $\Delta_g a$ on a general regular AES.

Let $d\mathrm{vol}_g$ be the positive Riemannian measure. The structure functions also determine the divergences of the arithmetic frame.

Proposition 2.1 (Structure functions and divergence). *The arithmetic frame satisfies*

$$\operatorname{div}_g X_u = -c_v, \qquad \operatorname{div}_g X_v = c_u. \tag{9}$$

Consequently, on compactly supported smooth scalar fields in the interior $M^\circ = M \setminus \partial M$, its formal adjoints are

$$X_u^* = -X_u + c_v, \qquad X_v^* = -X_v - c_u. \tag{10}$$

Here the star denotes a formal adjoint on $C_c^\infty(M^\circ) \subset L^2(M, d\mathrm{vol}_g)$, not a choice of a closed Hilbert-space realization.

Proof. For a positively oriented orthonormal frame on a surface, metric compatibility and vanishing torsion give

$$\langle \nabla_{X_v} X_u, X_v\rangle = -c_v, \qquad \langle \nabla_{X_u} X_v, X_u\rangle = c_u.$$

The remaining diagonal terms vanish because the frame has unit length. Taking the trace of $Y \mapsto \nabla_Y X_u$ and of $Y \mapsto \nabla_Y X_v$ proves (9). If X is a real vector field and $f, h \in C_c^\infty(M^\circ;\mathbb{C})$, the divergence theorem gives

$$\int_M (Xf)\,\overline{h}\, d\mathrm{vol}_g = -\int_M f\,\overline{(X + \operatorname{div}_g X)h}\, d\mathrm{vol}_g.$$

Substituting (9) yields (10). A coordinate-free sign check is recorded in Appendix A. □

2.2 The Laplace–Beltrami expression and energy

We retain the divergence-of-gradient convention

$$\Delta_g f = \operatorname{div}_g(\nabla f). \tag{11}$$

With this convention Δ_g is nonpositive in the L^2 quadratic-form sense on compactly supported functions.

Proposition 2.2 (Arithmetic-frame Laplacian). *For every $f \in C^\infty(M)$, the following pointwise identity holds:*

$$\boxed{\Delta_g f = X_u^2 f + X_v^2 f - c_v X_u f + c_u X_v f.} \tag{12}$$

On $C_c^\infty(M^\circ)$, the same differential expression has the formal energy factorization

$$-\Delta_g = X_u^* X_u + X_v^* X_v. \tag{13}$$

Proof. Because the frame is orthonormal,

$$\nabla f = (X_u f)X_u + (X_v f)X_v.$$

Using $\operatorname{div}(\phi X) = X\phi + \phi \operatorname{div} X$ and Proposition 2.1 gives (12). Substitution of the formal adjoints (10) gives (13) directly. □

In particular, the raw sum of squares

$$S_M := X_u^2 + X_v^2 \tag{14}$$

is not the Laplace–Beltrami operator unless $-c_v X_u + c_u X_v = 0$. The drift in (12) is fixed by $d\mathrm{vol}_g$; it is not an optional lower-order perturbation.

For later operator statements, define the densely defined nonnegative form

$$\mathcal{E}_0(f,h) := \int_M \big((X_u f)\overline{X_u h} + (X_v f)\overline{X_v h}\big)\, d\mathrm{vol}_g, \qquad f, h \in C_c^\infty(M^\circ; \mathbb{C}). \tag{15}$$

Its form norm is

$$\|f\|_{\mathcal{E}}^2 := \|f\|_{L^2(d\mathrm{vol}_g)}^2 + \mathcal{E}_0(f,f).$$

Definition 2.3 (Minimal energy domain and Dirichlet realization)**.** Let $H_0^1(M,g)$ denote the completion of $C_c^\infty(M^\circ)$ in the form norm above. The closure $\mathcal{E}$ of $\mathcal{E}_0$ is the Dirichlet energy form. We denote by $-\Delta_{g,D}$ the nonnegative self-adjoint operator associated with $\mathcal{E}$ by the representation theorem for closed forms.

This notation makes no assertion that the minimal differential operator is essentially self-adjoint. If M has boundary, it selects the Dirichlet form domain; Neumann, Robin, and mixed realizations require different form domains. If M is complete without boundary, standard completeness results may identify this realization with the unique self-adjoint closure, but no such conclusion is needed for the local identities below.

Proposition 2.4 (Weak and operator meanings)**.** *For $f \in C_c^\infty(M^\circ)$,*

$$\mathcal{E}_0(f,h) = \langle -\Delta_g f, h\rangle_{L^2(d\mathrm{vol}_g)} \qquad (h \in C_c^\infty(M^\circ)). \tag{16}$$

More generally, $f \in \mathrm{Dom}(-\Delta_{g,D})$ precisely when there is $k \in L^2(M, d\mathrm{vol}_g)$ such that

$$\mathcal{E}(f,h) = \langle k, h\rangle_{L^2(d\mathrm{vol}_g)} \quad \textit{for every } h \in H_0^1(M,g),$$

in which case $-\Delta_{g,D} f = k$.

Proof. Equation (16) follows from (13) and integration by parts; compact support removes all boundary terms. The second statement is the defining characterization of the operator represented by the closed form $\mathcal{E}$. □

Thus three levels will be kept separate throughout the paper:

1. pointwise identities of differential expressions on C^∞;
2. formal-adjoint identities on $C_c^\infty(M^\circ)$;
3. statements about a closed operator on a declared Hilbert-space domain.

Only the third level supports spectral terminology.

3 Arithmetic Cauchy–Riemann and Laplace operators

The preceding section fixed the measure and the real energy operator. We now complexify the arithmetic frame. Because M is two-dimensional, the resulting Cauchy–Riemann equation is the ordinary Riemann-surface equation written in the assignment-adapted frame; its factorization nevertheless exposes exactly where the frame bracket and the Laplace–Beltrami drift enter.

3.1 Compatible complex structure and unitary gauge

Let J be rotation by $+\pi/2$ in the oriented tangent planes. Thus

$$JX_u = X_v, \qquad JX_v = -X_u. \tag{17}$$

In real dimension two this almost-complex structure is integrable. No additional complex structure is being imposed on the regular AES.

Definition 3.1 (Surface arithmetic Cauchy–Riemann operators)**.** On complex-valued scalar fields define

$$\overline{\partial}_{\mathrm{AEG}} := \frac{1}{2}(X_u + iX_v), \qquad \partial_{\mathrm{AEG}} := \frac{1}{2}(X_u - iX_v). \tag{18}$$

These symbols denote first-order scalar coefficient operators. Intrinsically, if (η^u, η^v) is the coframe dual to (X_u, X_v), then

$$\bar{\partial}_M F = (\overline{\partial}_{\mathrm{AEG}} F)(\eta^u - i\eta^v). \tag{19}$$

The scalar coefficient in (19) depends on an orthonormal frame, while the $(0,1)$-form does not.

Proposition 3.2 ($U(1)$-gauge covariance)**.** *Let (Y_u, Y_v) be another positively oriented orthonormal frame on an open set $U \subset M$, related to the arithmetic frame by a smooth angle ϑ:*

$$Y_u = \cos\vartheta\, X_u + \sin\vartheta\, X_v, \qquad Y_v = -\sin\vartheta\, X_u + \cos\vartheta\, X_v. \tag{20}$$

Then

$$\frac{1}{2}(Y_u + iY_v) = e^{-i\vartheta}\overline{\partial}_{\mathrm{AEG}}. \tag{21}$$

Consequently, the equation $\overline{\partial}_{\mathrm{AEG}} F = 0$ is independent of the chosen oriented orthonormal frame. On the frame domain $U \subset M$, multiplication on $L^2(U, \mathrm{dvol}_g)$ by $e^{-i\vartheta}$ is unitary, so the quadratic operator $\overline{\partial}^*_{\mathrm{AEG}}\overline{\partial}_{\mathrm{AEG}}$ is gauge invariant on compactly supported test fields.

Proof. Expanding (20) gives

$$Y_u + iY_v = e^{-i\vartheta}(X_u + iX_v),$$

which proves (21). The dual coframe transforms by

$$\theta^u - i\theta^v = e^{i\vartheta}(\eta^u - i\eta^v),$$

so the two gauge factors cancel in (19). Finally, if U_ϑ is multiplication by $e^{-i\vartheta}$, then

$$(U_\vartheta\overline{\partial}_{\mathrm{AEG}})^*(U_\vartheta\overline{\partial}_{\mathrm{AEG}}) = \overline{\partial}^*_{\mathrm{AEG}} U^*_\vartheta U_\vartheta \overline{\partial}_{\mathrm{AEG}} = \overline{\partial}^*_{\mathrm{AEG}}\overline{\partial}_{\mathrm{AEG}}$$

as a formal-adjoint identity on C_c^∞. □

The arithmetic frame has a more specific gauge relation to the gradient frame selected by the assignment itself. This formula identifies exactly which part of the surface Cauchy–Riemann operator is AEG-specific.

Proposition 3.3 (Assignment-specific unitary gauge)**.** *Let*

$$q = (\mu^2 + \lambda^2 a^2)^{1/2}, \qquad E = \frac{\nabla a}{q}. \tag{22}$$

Then (E, JE) is a positively oriented orthonormal frame and

$$\boxed{X_u + iX_v = \frac{\mu + i\lambda a}{q}(E + iJE).} \tag{23}$$

The coefficient

$$\chi_a := \frac{\mu + i\lambda a}{q} \tag{24}$$

has unit modulus. Consequently, if

$$\bar{Z}_E := \frac{1}{2}(E + iJE),$$

then

$$\overline{\partial}_{\mathrm{AEG}} = \chi_a \bar{Z}_E, \qquad \ker \overline{\partial}_{\mathrm{AEG}} = \ker \bar{Z}_E \tag{25}$$

pointwise. Thus the arithmetic Cauchy–Riemann equation has exactly the same local solution sheaf as the underlying Riemann-surface $\bar{\partial}$-equation.

Proof. The eikonal identity (5) gives $|\nabla a| = q$, so (E, JE) is positively oriented and orthonormal. The canonical-frame formula imported from Paper I is

$$X_u = \frac{\mu}{q}E - \frac{\lambda a}{q}JE, \qquad X_v = \frac{\lambda a}{q}E + \frac{\mu}{q}JE.$$

Adding i times the second identity to the first gives

$$\begin{aligned} X_u + iX_v &= \frac{\mu + i\lambda a}{q}E + \frac{-\lambda a + i\mu}{q}JE \\ &= \frac{\mu + i\lambda a}{q}(E + iJE), \end{aligned}$$

because $i(\mu + i\lambda a) = i\mu - \lambda a$. Finally,

$$|\chi_a|^2 = \frac{\mu^2 + \lambda^2 a^2}{q^2} = 1.$$

The kernel statement follows because χ_a never vanishes. $\square$

3.2 Raw factorization and the metric drift

Let $S_M = X_u^2 + X_v^2$ as in (14). Direct multiplication of the two first-order expressions gives a factorization which does not yet involve a measure.

Proposition 3.4 (Raw frame factorization)**.** *On $C^\infty(M; \mathbb{C})$, pointwise as differential expressions,*

$$4\partial_{\mathrm{AEG}}\overline{\partial}_{\mathrm{AEG}} = S_M + i[X_u, X_v] = S_M + i(c_u X_u + c_v X_v), \tag{26}$$
$$4\overline{\partial}_{\mathrm{AEG}}\partial_{\mathrm{AEG}} = S_M - i[X_u, X_v] = S_M - i(c_u X_u + c_v X_v). \tag{27}$$

The Laplace–Beltrami expression is instead

$$\Delta_g = S_M - c_v X_u + c_u X_v. \tag{28}$$

Thus neither raw product in (26)– (27) is, by itself, the metric Laplacian.

Proof. Expand
$$(X_u - iX_v)(X_u + iX_v) = X_u^2 + X_v^2 + i(X_u X_v - X_v X_u)$$
and reverse the order for the second formula. Equation (28) is Proposition 2.2. $\square$

Proposition 3.5 (Exact drift factorizations)**.** *The raw products can be written directly in terms of the metric Laplacian:*
$$4\partial_{\mathrm{AEG}}\overline{\partial}_{\mathrm{AEG}} = \Delta_g + 2(c_v + ic_u)\overline{\partial}_{\mathrm{AEG}}, \tag{29}$$
$$4\overline{\partial}_{\mathrm{AEG}}\partial_{\mathrm{AEG}} = \Delta_g + 2(c_v - ic_u)\partial_{\mathrm{AEG}}. \tag{30}$$
These are pointwise identities of differential expressions on $C^\infty(M;\mathbb{C})$.

Proof. Using (28),
$$\begin{aligned}\Delta_g + 2(c_v + ic_u)\overline{\partial}_{\mathrm{AEG}} &= S_M - c_v X_u + c_u X_v + (c_v + ic_u)(X_u + iX_v)\\ &= S_M + i(c_u X_u + c_v X_v)\\ &= 4\partial_{\mathrm{AEG}}\overline{\partial}_{\mathrm{AEG}}\end{aligned}$$
by (26). Complex conjugation of the coefficients, or the same expansion using (27), proves the second identity. $\square$

The distinction between the bracket term and the metric drift is essential. The former records the non-coordinate arithmetic frame; the latter is forced by Riemannian divergence. They agree in the precise way needed on the kernel of the Cauchy–Riemann operator, as the next theorem shows.

3.3 Formal-adjoint factorization

Theorem 3.6 (Adjoint factorization of the surface Laplacian)**.** *On $C_c^\infty(M^\circ;\mathbb{C})$, with respect to $d\mathrm{vol}_g$,*
$$\overline{\partial}^*_{\mathrm{AEG}} = -\partial_{\mathrm{AEG}} + \frac{1}{2}(c_v + ic_u), \tag{31}$$
$$\partial^*_{\mathrm{AEG}} = -\overline{\partial}_{\mathrm{AEG}} + \frac{1}{2}(c_v - ic_u). \tag{32}$$
Consequently,
$$\boxed{4\overline{\partial}^*_{\mathrm{AEG}}\overline{\partial}_{\mathrm{AEG}} = 4\partial^*_{\mathrm{AEG}}\partial_{\mathrm{AEG}} = -\Delta_g.} \tag{33}$$
These are formal differential identities on the common test domain; no boundary condition or self-adjoint domain is implicit in the star notation.

Proof. Using (10) and conjugating scalar coefficients when taking the adjoint,
$$\begin{aligned}\overline{\partial}^*_{\mathrm{AEG}} &= \frac{1}{2}(X_u^* - iX_v^*) = \frac{1}{2}(-X_u + c_v + iX_v + ic_u)\\ &= -\partial_{\mathrm{AEG}} + \frac{1}{2}(c_v + ic_u),\end{aligned}$$
which proves (31); the other formula is its conjugate counterpart. Now use (26):
$$\begin{aligned}4\overline{\partial}^*_{\mathrm{AEG}}\overline{\partial}_{\mathrm{AEG}} &= -4\partial_{\mathrm{AEG}}\overline{\partial}_{\mathrm{AEG}} + 2(c_v + ic_u)\overline{\partial}_{\mathrm{AEG}}\\ &= -S_M - i(c_u X_u + c_v X_v) + (c_v + ic_u)(X_u + iX_v)\\ &= -S_M + c_v X_u - c_u X_v\\ &= -\Delta_g.\end{aligned}$$
The calculation for $4\partial^*_{\mathrm{AEG}}\partial_{\mathrm{AEG}}$ is analogous. Appendix A records both expansions term by term. $\square$

The formal identity has a closed-operator consequence once the domain is stated.

Corollary 3.7 (Dirichlet realization by the CR energy)**.** *Let* $\overline{\overline{\partial}_{\mathrm{AEG}0}}$ *be the graph closure in* $L^2(M, d\mathrm{vol}_g)$ *of* $\overline{\partial}_{\mathrm{AEG}}$ *initially defined on* $C_c^\infty(M^\circ; \mathbb{C})$*. Then*

$$-\Delta_{g,D} = 4(\overline{\overline{\partial}_{\mathrm{AEG}0}})^* \overline{\overline{\partial}_{\mathrm{AEG}0}}. \tag{34}$$

The equality is an equality of nonnegative self-adjoint operators represented by the same closed quadratic form, not merely an equality of pointwise formulas.

Proof. For $f \in C_c^\infty(M^\circ; \mathbb{C})$, Theorem 3.6 gives

$$4\|\overline{\partial}_{\mathrm{AEG}} f\|^2_{L^2(d\mathrm{vol}_g)} = \langle -\Delta_g f, f\rangle = \mathcal{E}_0(f, f). \tag{35}$$

Moreover,

$$\|f\|^2_{L^2(d\mathrm{vol}_g)} + 4\|\overline{\partial}_{\mathrm{AEG}} f\|^2_{L^2(d\mathrm{vol}_g)} = \|f\|^2_{\mathcal{E}}.$$

Thus $\overline{\partial}_{\mathrm{AEG}}$ is closable with graph domain $H_0^1(M, g)$, and the closure of the form on the left of (35) is exactly the Dirichlet form $\mathcal{E}$ from Definition 2.3. The uniqueness of the self-adjoint operator represented by a closed nonnegative form proves (34). □

3.4 Arithmetic holomorphicity and harmonicity

Definition 3.8 (Arithmetic holomorphic field on a regular AES)**.** Let $\Omega \subset M$ be open. A field $F \in C^1(\Omega; \mathbb{C})$ is *arithmetic holomorphic* if

$$\overline{\partial}_{\mathrm{AEG}} F = 0 \quad \text{on } \Omega. \tag{36}$$

Equivalently, if $F = f + ih$, then

$$X_u f = X_v h, \qquad X_v f = -X_u h. \tag{37}$$

By Proposition 3.2, this definition is independent of the arithmetic frame used to write the coefficient equation. By (19), it is exactly ordinary holomorphicity for the Riemann surface (M, J).

Proposition 3.9 (Elementary holomorphic calculus)**.** *Let $F = f + ih$ be arithmetic holomorphic on an open set $\Omega \subset M$. Then the following statements hold.*

1. *If Φ is classically holomorphic on a complex neighborhood of $F(\Omega)$, then $\Phi \circ F$ is arithmetic holomorphic.*

2. *The two surface derivative vectors are orthogonal and have equal length:*

$$(X_u f)(X_v f) + (X_u h)(X_v h) = 0, \tag{38}$$
$$(X_u f)^2 + (X_u h)^2 = (X_v f)^2 + (X_v h)^2. \tag{39}$$

   The oriented Jacobian satisfies

$$J_g(F) := (X_u f)(X_v h) - (X_v f)(X_u h) = (X_u f)^2 + (X_v f)^2 \geq 0. \tag{40}$$

3. *If Ω is connected and $F = H \circ a$ for a C^1 function H on an interval containing $a(\Omega)$, then F is constant.*

Proof. The real vector fields (X_u, X_v) obey the ordinary chain rule. Since Φ is holomorphic, its complex derivative gives

$$\overline{\partial}_{\mathrm{AEG}}(\Phi \circ F) = \Phi'(F)\overline{\partial}_{\mathrm{AEG}} F = 0,$$

which proves the first statement. The real Cauchy–Riemann system (37) says

$$X_u f = X_v h, \qquad X_u h = -X_v f.$$

Substitution proves (38)– (40).

For the final statement, the assignment derivative rules (6) give

$$2\overline{\partial}_{\mathrm{AEG}}(H \circ a) = (\mu + i\lambda a)H'(a). \tag{41}$$

The factor $\mu + i\lambda a$ never vanishes for real a, because $\mu \neq 0$. Hence $H'(a) = 0$ throughout Ω, so $d(H \circ a) = 0$. Connectedness makes F constant. $\square$

Theorem 3.10 (Holomorphic fields are Laplace–Beltrami harmonic)**.** *If $F \in C^2(\Omega; \mathbb{C})$ is arithmetic holomorphic, then*

$$\Delta_g F = 0 \quad \textit{pointwise on } \Omega. \tag{42}$$

Hence the real and imaginary parts of F are real Laplace–Beltrami harmonic functions.

Proof. The formal-adjoint formula (31) is also an identity of local differential expressions. Therefore

$$-\Delta_g F = 4\overline{\partial}^*_{\mathrm{AEG}}(\overline{\partial}_{\mathrm{AEG}} F) = 0$$

pointwise. Taking real and imaginary parts proves the final statement.

Equivalently, the raw factorization gives

$$0 = 4\partial_{\mathrm{AEG}}\overline{\partial}_{\mathrm{AEG}} F = S_M F + i(c_u X_u + c_v X_v)F.$$

On the Cauchy–Riemann kernel, $X_u F = -iX_v F$, and hence

$$i(c_u X_u + c_v X_v)F = (-c_v X_u + c_u X_v)F.$$

The right-hand side converts $S_M F$ exactly into $\Delta_g F$ through (28). $\square$

Remark 3.11 (Local equation versus Hilbert-space kernel). Theorem 3.10 is local and pointwise; it does not require $F \in L^2$. By contrast, a statement such as $F \in \ker(-\Delta_{g,D})$ requires $F \in \mathrm{Dom}(-\Delta_{g,D})$ and the chosen Dirichlet boundary condition. Many holomorphic fields used later for ideal-boundary data are not square-integrable with respect to hyperbolic volume, so the two assertions must not be interchanged.

4 The contact-CR analytic layer

The analytic object considered in this section is the three-dimensional contact state space of Paper I. It is not a regular AES surface. We use D_u, D_v for its horizontal lifts and reserve X_u, X_v for the canonical frame of a two-dimensional regular AES. No identification of these frames or of their associated Laplacians is made here.

Assume throughout that $\mu, \lambda \in \mathbb{R}$ and $\mu\lambda \neq 0$, and put

$$\mathcal{C} = \mathbb{R}^3_{(u,v,a)}, \qquad \alpha = da - \mu\, du - \lambda a\, dv,$$

$$\mathcal{D} = \ker \alpha, \qquad D_u = \partial_u + \mu\partial_a, \qquad D_v = \partial_v + \lambda a \partial_a, \qquad T = \partial_a. \tag{43}$$

The conventions agree with Paper I, in particular

$$[D_u, D_v] = \mu\lambda T. \tag{44}$$

4.1 Horizontal metric, complex structure, and Frobenius obstruction

The contact form alone does not select the analytic normalization below. We make the additional choice explicit.

Definition 4.1 (Normalized arithmetic contact data)**.** The normalized arithmetic contact data consist of $\mathcal{D}$, the horizontal metric h, and the endomorphism J defined by

$$h(D_i, D_j) = \delta_{ij}, \qquad JD_u = D_v, \qquad JD_v = -D_u. \tag{45}$$

The ordered frame (D_u, D_v) fixes the horizontal orientation. The associated positive contact measure is

$$d\nu_{\mathrm{c}} := |\alpha \wedge d\alpha| = |\mu\lambda|\, du\, dv\, da, \tag{46}$$

where the last expression denotes a positive density.

Thus J is compatible with the chosen horizontal metric and orientation. We do not claim that it is selected by the contact distribution without this normalization, or that it is invariant under arbitrary contactomorphisms.

Proposition 4.2 (Frobenius nonintegrability)**.** *The distribution $\mathcal{D}$ is bracket generating of step two and is not Frobenius integrable. At every $p \in \mathcal{C}$,*

$$\mathcal{D}_p + [\,\mathcal{D}, \mathcal{D}\,]_p = T_p\mathcal{C}, \qquad \alpha([D_u, D_v]) = \mu\lambda \neq 0.$$

Consequently no immersed surface in $\mathcal{C}$ can have tangent bundle equal to $\mathcal{D}$ on a nonempty open set.

Proof. The horizontal fields are independent and project to ∂_u, ∂_v. Equation (44) supplies the vertical field T, so $D_u, D_v, [D_u, D_v]$ span every tangent space. Since $\alpha(T) = 1$, the bracket is not horizontal. The Frobenius theorem gives the remaining assertions. □

This is the sharp reason that the contact branch cannot be regarded as another coordinate presentation of the surface branch in Sections 2–3.

4.2 Adjoints and the variational sub-Laplacian

Formal-adjoint statements are initially made on $C_c^\infty(\mathcal{C}) \subset L^2(\mathcal{C}, d\nu_{\mathrm{c}})$, with the inner product complex linear in its first entry.

Proposition 4.3 (Contact-volume divergences and adjoints)**.** *With respect to $d\nu_{\mathrm{c}}$,*

$$\operatorname{div}_{\nu_{\mathrm{c}}} D_u = 0, \qquad \operatorname{div}_{\nu_{\mathrm{c}}} D_v = \lambda. \tag{47}$$

Consequently,

$$D_u^* = -D_u, \qquad D_v^* = -D_v - \lambda. \tag{48}$$

Proof. The density in (46) is a nonzero constant multiple of Lebesgue measure. The coordinate divergence of $D_u = \partial_u + \mu\partial_a$ is zero, whereas that of $D_v = \partial_v + \lambda a \partial_a$ is $\partial_a(\lambda a) = \lambda$. Integration by parts gives $Y^* = -Y - \operatorname{div} Y$ for a real vector field Y. □

Definition 4.4 (Contact energy and variational sub-Laplacian)**.** For $f, g \in C_c^\infty(\mathcal{C})$, set

$$\mathcal{E}_{\mathrm{c}}(f, g) = \int_{\mathcal{C}} (D_u f\, \overline{D_u g} + D_v f\, \overline{D_v g})\, d\nu_{\mathrm{c}}. \tag{49}$$

The divergence-of-horizontal-gradient operator and its nonnegative energy operator are

$$\Delta_{\mathcal{C}} := D_u^2 + D_v^2 + \lambda D_v, \tag{50}$$

$$A_{\mathcal{C}} := -\Delta_{\mathcal{C}} = D_u^* D_u + D_v^* D_v. \tag{51}$$

Proposition 4.5 (Energy identity and Friedrichs realization)**.** *The form $\mathcal{E}_{\mathrm{c}}$ is densely defined, nonnegative, and closable in $L^2(\mathcal{C}, d\nu_{\mathrm{c}})$. Its closure determines a unique nonnegative self-adjoint Friedrichs operator $A_{\mathcal{C}}^{\mathrm{F}}$. On $C_c^\infty(\mathcal{C})$ its differential expression is $A_{\mathcal{C}}$, and*

$$\langle A_{\mathcal{C}} f, f\rangle = \mathcal{E}_{\mathrm{c}}(f,f) = \|D_u f\|_2^2 + \|D_v f\|_2^2. \tag{52}$$

Proof. Define

$$\mathsf{D} : C_c^\infty(\mathcal{C}) \longrightarrow L^2(d\nu_{\mathrm{c}})^2, \qquad \mathsf{D}f = (D_u f, D_v f).$$

Proposition 4.3 shows that $\mathrm{Dom}(\mathsf{D}^*)$ contains the dense subspace $C_c^\infty(\mathcal{C})^2$. Hence D is closable, and so is the quadratic form $\|\mathsf{D}f\|_2^2$. The representation theorem for closed nonnegative forms gives the Friedrichs operator. Integration by parts gives

$$\mathcal{E}_{\mathrm{c}}(f,g) = \langle (D_u^* D_u + D_v^* D_v) f, g\rangle = \langle A_{\mathcal{C}} f, g\rangle,$$

which proves the differential expression and the energy identity. □

Theorem 4.6 (Hörmander hypoellipticity)**.** *The differential expressions $A_{\mathcal{C}}$ and $\Delta_{\mathcal{C}}$ are hypoelliptic. If $A_{\mathcal{C}} f$ is smooth on an open set in the distributional sense, then f is smooth there.*

Proof. Proposition 4.2 verifies Hörmander's bracket condition for D_u, D_v. The second-order part is their sum of squares and the remaining term is first order, so the sum-of-squares theorem applies [3]. □

Remark 4.7*.* This is a local regularity statement. It does not assert essential self-adjointness on $C_c^\infty(\mathcal{C})$, identify the Friedrichs spectrum, or solve a boundary problem on an arbitrary contact domain.

4.3 Raw and adjoint CR factorizations

Definition 4.8 (Contact CR operators)**.** Set

$$\overline{L} = \frac{1}{2}(D_u + iD_v), \qquad L = \frac{1}{2}(D_u - iD_v), \qquad Q_{\mathcal{C}} = D_u^2 + D_v^2. \tag{53}$$

A smooth complex field F is contact arithmetic CR when $\overline{L}F = 0$.

Proposition 4.9 (Raw curvature-twisted factorization)**.** *On smooth complex fields,*

$$4L\overline{L} = Q_{\mathcal{C}} + i\mu\lambda T, \tag{54}$$
$$4\overline{L}L = Q_{\mathcal{C}} - i\mu\lambda T. \tag{55}$$

Every contact arithmetic CR field therefore satisfies

$$(\Delta_{\mathcal{C}} - \lambda D_v + i\mu\lambda T)F = 0. \tag{56}$$

Proof. Expansion gives

$$4L\overline{L} = (D_u - iD_v)(D_u + iD_v) = D_u^2 + D_v^2 + i[D_u, D_v].$$

This proves the first identity; reversing the order proves the second. For a CR field, apply L and use $Q_{\mathcal{C}} = \Delta_{\mathcal{C}} - \lambda D_v$. □

Proposition 4.10 (Contact-volume adjoint factorizations)**.** *As formal identities on $C_c^\infty(\mathcal{C}) \subset L^2(d\nu_{\mathrm{c}})$,*

$$\overline{L}^* = \frac{1}{2}(-D_u + iD_v + i\lambda), \qquad L^* = \frac{1}{2}(-D_u - iD_v - i\lambda), \tag{57}$$

and

$$4\overline{L}^*\overline{L} = A_{\mathcal{C}} + i\lambda D_u - i\mu\lambda T, \tag{58}$$

$$4L^*L = A_{\mathcal{C}} - i\lambda D_u + i\mu\lambda T. \tag{59}$$

The Reeb field of α is

$$\mathcal{R} = -\mu^{-1}\partial_u, \tag{60}$$

and the same identities therefore take the intrinsic contact form

$$4\overline{L}^*\overline{L} = A_{\mathcal{C}} - i\mu\lambda\mathcal{R}, \tag{61}$$

$$4L^*L = A_{\mathcal{C}} + i\mu\lambda\mathcal{R}. \tag{62}$$

Proof. The adjoints follow from (48), with scalar coefficients conjugated. Expanding the first product,

$$\begin{aligned} 4\overline{L}^*\overline{L} &= (-D_u + iD_v + i\lambda)(D_u + iD_v) \\ &= -D_u^2 - D_v^2 - \lambda D_v + i\lambda D_u - i[D_u, D_v] \\ &= A_{\mathcal{C}} + i\lambda D_u - i\mu\lambda T. \end{aligned}$$

The second calculation is analogous and gives the stated signs. Finally, $d\alpha = -\lambda\, da \wedge dv$ has kernel spanned by ∂_u, and $\alpha(-\mu^{-1}\partial_u) = 1$; hence $\mathcal{R} = -\mu^{-1}\partial_u$. Substitution of $D_u = \partial_u + \mu T$ into the two preceding formulas gives their Reeb forms. □

The raw identities display vertical curvature. The adjoint identities also contain the measure-dependent drift required by the contact energy. In particular, every smooth contact arithmetic CR field satisfies the pointwise equation

$$(A_{\mathcal{C}} - i\mu\lambda\mathcal{R})F = 0, \qquad \text{equivalently} \qquad (\Delta_{\mathcal{C}} + i\mu\lambda\mathcal{R})F = 0. \tag{63}$$

This differential consequence does not assert that F belongs to the L^2-domain of the Friedrichs operator.

4.4 The balanced measure and its unitary gauge

Definition 4.11 (Balanced horizontal measure)**.** Define

$$d\nu_{\mathrm{b}} = e^{-\lambda v}\, d\nu_{\mathrm{c}}. \tag{64}$$

Multiplication by

$$Uf = e^{-\lambda v/2} f \tag{65}$$

defines a unitary map from $L^2(d\nu_{\mathrm{b}})$ to $L^2(d\nu_{\mathrm{c}})$.

Proposition 4.12 (Two-measure gauge identities)**.** *With respect to $d\nu_{\mathrm{b}}$, both horizontal fields are divergence free. Thus $D_u^{*,\mathrm{b}} = -D_u$, $D_v^{*,\mathrm{b}} = -D_v$, and $-Q_{\mathcal{C}}$ is the balanced-measure energy expression. Moreover,*

$$UD_uU^{-1} = D_u, \tag{66}$$

$$UD_vU^{-1} = D_v + \frac{\lambda}{2}, \tag{67}$$

$$U(-Q_{\mathcal{C}})U^{-1} = A_{\mathcal{C}} - \frac{\lambda^2}{4}, \tag{68}$$

$$U\overline{L}U^{-1} = \overline{L} + \frac{i\lambda}{4}, \qquad ULU^{-1} = L - \frac{i\lambda}{4}. \tag{69}$$

Proof. For $e^{\phi}d\nu_{\mathrm{c}}$, $\operatorname{div}_{e^{\phi}\nu_{\mathrm{c}}} Y = \operatorname{div}_{\nu_{\mathrm{c}}} Y + Y\phi$. With $\phi = -\lambda v$, one has $D_u\phi = 0$ and $D_v\phi = -\lambda$, which cancels (47). Unitarity follows directly from

$$\int |Uf|^2 d\nu_{\mathrm{c}} = \int |f|^2 e^{-\lambda v} d\nu_{\mathrm{c}}.$$

Since $D_uv = 0$ and $D_vv = 1$, differentiating $U^{-1}f = e^{\lambda v/2}f$ proves the first two conjugation formulas. Hence

$$U(-Q_{\mathcal{C}})U^{-1} = -D_u^2 - \left(D_v + \frac{\lambda}{2}\right)^2 = A_{\mathcal{C}} - \frac{\lambda^2}{4}.$$

The CR formulas follow from their definitions. ☐

Warning 4.13. The balanced measure is an arithmetic-coordinate gauge, not the positive contact measure. Every statement about adjoints, spectra, or L^2-membership must name the measure used.

4.5 Logarithmic CR first integrals and modes

Choose a holomorphic branch Log_μ on

$$\Omega_\mu = \{w \in \mathbb{C} : \operatorname{sgn}(\mu)\Re w > 0\}.$$

The line $\mu + i\lambda\mathbb{R}$ lies in this half-plane.

Proposition 4.14 (Two CR first integrals)**.** *The fields*

$$z = u + iv, \qquad \zeta = u + \frac{i}{\lambda}\mathrm{Log}_\mu(\mu + i\lambda a) \tag{70}$$

satisfy $\overline{L}z = \overline{L}\zeta = 0$. The map

$$\Psi : \mathcal{C} \longrightarrow \mathbb{C}^2, \qquad \Psi = (z, \zeta), \tag{71}$$

has real rank three everywhere.

Proof. One has $D_uz = 1$, $D_vz = i$, and hence $\overline{L}z = \frac{1}{2}(1 + i^2) = 0$. Further,

$$\partial_a\zeta = -\frac{1}{\mu + i\lambda a}, \quad D_u\zeta = \frac{i\lambda a}{\mu + i\lambda a}, \quad D_v\zeta = -\frac{\lambda a}{\mu + i\lambda a},$$

so $D_u\zeta + iD_v\zeta = 0$. Finally,

$$dz = du + i\,dv, \qquad d\zeta = du - \frac{da}{\mu + i\lambda a}.$$

If a real tangent vector is killed by both differentials, the first forces its u, v components to vanish and the second forces its a component to vanish. Thus $d\Psi$ is injective. ☐

Corollary 4.15 (Explicit contact CR modes)**.** *If Φ is holomorphic on a neighborhood of $\Psi(\mathcal{C})$, then $\Phi(z, \zeta)$ is contact arithmetic CR. In particular, for $\rho, \sigma \in \mathbb{C}$,*

$$F_{\rho,\sigma} = e^{\rho z + \sigma\zeta} = e^{(\rho+\sigma)u + i\rho v}(\mu + i\lambda a)^{i\sigma/\lambda} \tag{72}$$

is a smooth CR field in the chosen logarithmic branch and depends genuinely on a when $\sigma \neq 0$.

Proof. The chain rule and Proposition 4.14 give

$$\overline{L}\Phi(z, \zeta) = \Phi_1 \overline{L} z + \Phi_2 \overline{L}\zeta = 0.$$

The displayed modes use $\Phi(w_1, w_2) = e^{\rho w_1 + \sigma w_2}$. □

No square-integrability, orthogonality, spanning, or completeness assertion is made for these modes.

4.6 The filtered affine–Appell module

Let $r = e^{\lambda v}$, set

$$B_n = r^{-n} a^n, \qquad n \geq 0, \tag{73}$$

and let $\mathcal{A} = C^\infty(\mathbb{R}^2_{(u,v)}, \mathbb{C})$, pulled back to $\mathcal{C}$. Define

$$\mathcal{M}_N = \left\{ \sum_{n=0}^{N} P_n(u, v) B_n : P_n \in \mathcal{A} \right\}. \tag{74}$$

We also set $\mathcal{M}_{-1} = \{0\}$.

Proposition 4.16 (Filtered affine–Appell stability)**.** *The modules $\mathcal{M}_N$ form an increasing filtration and $\mathcal{M}_N \mathcal{M}_M \subseteq \mathcal{M}_{N+M}$. For $P \in \mathcal{A}$, with $B_{-1} = 0$,*

$$D_u(PB_n) = (\partial_u P)B_n + \mu n r^{-1} P B_{n-1}, \tag{75}$$

$$D_v(PB_n) = (\partial_v P)B_n, \tag{76}$$

$$T(PB_n) = n r^{-1} P B_{n-1}. \tag{77}$$

Each $\mathcal{M}_N$ is stable under $D_u, D_v, \overline{L}, L, Q_{\mathcal{C}}, \Delta_{\mathcal{C}}$, and $A_{\mathcal{C}}$, while $T\mathcal{M}_N \subseteq \mathcal{M}_{N-1}$.

Proof. The product assertion follows from $B_n B_m = B_{n+m}$. Direct differentiation gives

$$D_u B_n = \mu n r^{-1} B_{n-1}, \qquad D_v B_n = 0, \qquad T B_n = n r^{-1} B_{n-1}.$$

Since r^{-1} is smooth in v, the coefficients remain in $\mathcal{A}$. First-order stability follows by linear combination and second-order stability by iteration. □

Remark 4.17. The term *module* is essential. The finite-dimensional complex span of $B_0, \ldots, B_N$ is not stable under D_u, because r^{-1} appears in (75). No density or completeness claim is made. The finite upward-sweep antiderivative is proved in Appendix B.

5 Holomorphic pullbacks and the cylindrical AES

The intrinsic complex structure of a regular AES makes holomorphic pullback a natural source of new regular examples. This section records the construction in the form needed by Paper III. Every statement below is confined to the locus on which the relevant holomorphic map is a local biholomorphism. No assertion is made about extending through a critical or branch set, or through a zero or pole of the cylindrical target map.

5.1 A planar harmonic base AES

Fix $\phi \in \mathbb{R}$, write $w = \xi + i\eta$, and define

$$A_\phi(w) := \operatorname{Im}(e^{-i\phi} w), \qquad Q_\phi(w)^2 := \mu^2 + \lambda^2 A_\phi(w)^2, \tag{78}$$

where $\mu \neq 0$ and $\lambda \in \mathbb{R}$. On $\mathbb{C}$, set

$$h_\phi := \frac{|dw|^2}{Q_\phi(w)^2}. \tag{79}$$

The phase merely rotates the Euclidean coordinate in which the assignment is the second coordinate, but retaining it makes the zero direction explicit.

Proposition 5.1 (Planar harmonic AES and curvature law)**.** *The tuple* $(\mathbb{C}, h_\phi, A_\phi; \mu, \lambda)$ *is a regular AES and*

$$|\nabla A_\phi|^2_{h_\phi} = \mu^2 + \lambda^2 A_\phi^2, \tag{80}$$

$$\Delta_{h_\phi} A_\phi = 0, \tag{81}$$

$$K_{h_\phi} = \lambda^2 \frac{\mu^2 - \lambda^2 A_\phi^2}{\mu^2 + \lambda^2 A_\phi^2}. \tag{82}$$

In particular, the AES equation by itself neither determines the curvature nor forces the assignment to satisfy the pointwise equation $\Delta_g a = 2\lambda^2 a$ *of the basic hyperbolic model.*

Proof. After the Euclidean rotation $u + iv = e^{-i\phi} w$, one has $A_\phi = v$ and

$$h_\phi = e^{2\rho(v)}(du^2 + dv^2), \qquad \rho(v) = -\frac{1}{2}\log(\mu^2 + \lambda^2 v^2).$$

Since the inverse metric is $e^{-2\rho}$ times the Euclidean inverse metric,

$$|\nabla v|^2_{h_\phi} = e^{-2\rho} = \mu^2 + \lambda^2 v^2.$$

In real dimension two the scalar Laplacian obeys $\Delta_{e^{2\rho} g_0} f = e^{-2\rho}\Delta_{g_0} f$; hence $\Delta_{h_\phi} v = 0$. Finally,

$$\rho''(v) = -\lambda^2 \frac{\mu^2 - \lambda^2 v^2}{(\mu^2 + \lambda^2 v^2)^2},$$

and the conformal curvature formula $K = -e^{-2\rho}\Delta_{g_0}\rho$ gives (82). $\square$

The zero locus is the Euclidean line

$$Z(A_\phi) = e^{i\phi}\mathbb{R}. \tag{83}$$

It is regular because $|\nabla A_\phi|_{h_\phi} = |\mu|$ there. The next theorem shows that its inverse images give all zero curves produced by this base model on the regular locus.

5.2 Locally biholomorphic pullback

Theorem 5.2 (Holomorphic pullback of the planar base)**.** *Let Σ be a Riemann surface, $U \subset \Sigma$ open, and $F : U \to \mathbb{C}$ holomorphic with $dF_p \neq 0$ for every $p \in U$. Define*

$$a_F := A_\phi \circ F, \qquad g_F := F^* h_\phi. \tag{84}$$

Then $(U, g_F, a_F; \mu, \lambda)$ is a regular AES. Moreover,

$$\Delta_{g_F} a_F = 0, \tag{85}$$

$$K_{g_F} = \lambda^2 \frac{\mu^2 - \lambda^2 a_F^2}{\mu^2 + \lambda^2 a_F^2}. \tag{86}$$

If z is a local complex coordinate on U, then

$$g_F = \frac{|F'(z)|^2}{\mu^2 + \lambda^2 a_F(z)^2} |dz|^2, \qquad Z(a_F) = F^{-1}(e^{i\phi}\mathbb{R}). \tag{87}$$

Finally, every holomorphic field Ψ on an open neighborhood of $F(U)$ pulls back to the arithmetic holomorphic field $\Psi \circ F$ on U.

Proof. The nonvanishing of dF makes $F : (U, g_F) \to (\mathbb{C}, h_\phi)$ a local oriented isometry. The norm of the gradient, the scalar Laplacian, and Gaussian curvature are preserved under local isometries, so Proposition 5.1 gives the AES equation and (85)–(86). The coordinate expression follows directly from $F^*|dw|^2 = |F'(z)|^2|dz|^2$, and the zero-set identity follows from (83). Since F is holomorphic for the given Riemann-surface structures, so is $\Psi \circ F$; by Definition 3.8, this is arithmetic holomorphicity for g_F. □

Warning 5.3 (The critical set is not part of the theorem). If $F'(p) = 0$, formula (87) defines a degenerate quadratic form at p, not a Riemannian metric. If, in addition, $A_\phi(F(p)) = 0$, the inverse image of the target zero line may have several local branches. Such a point cannot be regular: the AES equation would require $|\nabla a_F| = |\mu|$ on its zero set. Branched pullbacks and their zero-graph normal forms belong to the singular theory of Paper III.

5.3 The complete cylindrical AES

The planar construction is adapted to inverse images of a line. For inverse images of a circle, the appropriate regular base is the logarithmic cylinder. Let W denote the standard coordinate on $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$, and put

$$s(W) := \log |W|, \qquad g_{\text{cyl}} := \left| \frac{dW}{W} \right|^2. \tag{88}$$

Define

$$a_{\text{cyl}}(W) := \begin{cases} \dfrac{\mu}{\lambda} \sinh(\lambda s(W)), & \lambda \neq 0, \\ \mu s(W), & \lambda = 0. \end{cases} \tag{89}$$

Proposition 5.4 (Cylindrical AES)**.** *The tuple $(\mathbb{C}^*, g_{\text{cyl}}, a_{\text{cyl}}; \mu, \lambda)$ is a complete regular AES. It is flat and satisfies*

$$|\nabla a_{\text{cyl}}|^2_{g_{\text{cyl}}} = \mu^2 + \lambda^2 a_{\text{cyl}}^2, \tag{90}$$

$$\Delta_{g_{\text{cyl}}} a_{\text{cyl}} = \lambda^2 a_{\text{cyl}}, \tag{91}$$

$$Z(a_{\text{cyl}}) = \{|W| = 1\}. \tag{92}$$

For $\lambda = 0$, equation (91) has the stated limiting meaning $\Delta a_{\text{cyl}} = 0$.

Proof. Locally choose a logarithm $\zeta = \log W = s + i\theta$. Then

$$g_{\mathrm{cyl}} = |d\zeta|^2 = ds^2 + d\theta^2.$$

The universal covering map identifies this metric with the quotient of the Euclidean plane by $\theta \mapsto \theta + 2\pi$, so it is complete and flat. For $\lambda \neq 0$,

$$\frac{da_{\mathrm{cyl}}}{ds} = \mu \cosh(\lambda s), \qquad \frac{d^2 a_{\mathrm{cyl}}}{ds^2} = \lambda^2 a_{\mathrm{cyl}}.$$

The identity $\cosh^2 t = 1 + \sinh^2 t$ proves (90), while the flat-cylinder Laplacian proves (91). The formulas for $\lambda = 0$ follow directly. Since $\sinh t = 0$ only at $t = 0$, the zero set is $s = 0$, equivalently $|W| = 1$. $\square$

As in Theorem 5.2, any holomorphic local biholomorphism $W : U \to \mathbb{C}^*$ gives a regular pullback AES

$$g_W = \left|\frac{dW}{W}\right|^2, \qquad a_W = \begin{cases} \dfrac{\mu}{\lambda} \sinh(\lambda \log |W|), & \lambda \neq 0, \\ \mu \log |W|, & \lambda = 0, \end{cases} \tag{93}$$

with regular zero locus

$$Z(a_W) = W^{-1}(S^1). \tag{94}$$

On this pullback one has $K_{g_W} = 0$ and $\Delta_{g_W} a_W = \lambda^2 a_W$.

5.4 Automorphic descent and the Paper III interface

The preceding formula is useful when W is constructed from an automorphic or meromorphic function. The following elementary descent statement isolates what is already regular analysis from what remains singular geometry.

Proposition 5.5 (Unitary automorphy and descent)**.** *Let a group Γ act biholomorphically on a Riemann surface U, and let $W : U \to \mathbb{C}^*$ be a holomorphic local biholomorphism. Suppose there is a character $\chi : \Gamma \to S^1$ such that*

$$W(\gamma p) = \chi(\gamma) W(p). \tag{95}$$

Then g_W, a_W, and $Z(a_W)$ are Γ-invariant and descend to every smooth quotient on which the action is free and properly discontinuous.

More generally, suppose the target transformations define an action and have the form

$$W(\gamma p) = \chi_\gamma W(p)^{\varepsilon(\gamma)}, \qquad |\chi_\gamma| = 1, \qquad \varepsilon(\gamma) \in \{1, -1\}. \tag{96}$$

Then $\varepsilon : \Gamma \to \{1, -1\}$ is a homomorphism, g_W and $Z(a_W)$ are invariant, and a_W is multiplied by $\varepsilon(\gamma)$. It therefore descends as a scalar only on the sign-preserving subgroup; on the full quotient it is naturally a section of the corresponding real sign line bundle.

Proof. Multiplication by a unit complex number preserves $\log |W|$ and dW/W. Inversion sends these quantities to $-\log |W|$ and $-dW/W$, respectively. Thus the metric and the unit-circle preimage are unchanged, whereas the odd function in (89) changes sign. Standard quotient descent then proves the statement. $\square$

Remark 5.6 (Converting an automorphic interval into the zero circle). Let β be a meromorphic automorphic function. On an open sheet avoiding its branch values, choose R and define W by

$$R^2 = \frac{\beta}{1-\beta}, \qquad W = \frac{R-i}{R+i}. \tag{97}$$

Where these expressions are finite and locally biholomorphic,

$$|W| = 1 \quad \Longleftrightarrow \quad R \in \mathbb{R} \cup \{\infty\} \quad \Longleftrightarrow \quad \beta \in [0,1]. \tag{98}$$

Changing the square-root sheet sends R to $-R$ and W to W^{-1}. Thus the cylindrical metric and zero locus are independent of that sheet change, whereas the assignment changes sign. This observation supplies the exact regular analytic interface for a normalized Hauptmodul or similar automorphic coordinate. It does not assert that $\beta^{-1}([0,1])$ has any particular global graph type; that conclusion requires separate ramification and tessellation data.

For a meromorphic or algebraic automorphic construction, define its Paper II regular locus to be the union of open sheets on which

$$0 < |W| < \infty, \qquad dW \neq 0, \tag{99}$$

and on which a required algebraic branch has been chosen. Equations (93)–(94) apply on that locus without change. Zeros and poles of W are excluded because $\log |W|$ is not a finite smooth assignment there. Critical points are excluded because the pulled-back metric degenerates, and algebraic branch points require a separate local uniformization. These excluded sets are among the natural loci at which inverse images of S^1 may meet or branch, and at which they may accumulate. Boundary escape and nonproperness can also cause accumulation without appearing in this local list. Paper II exports the regular formula and the descent criterion; Paper III owns the completion, local branching, singular zero graph, and parameterized tube topology.

6 The basic hyperbolic analytic model

We now specialize the surface theory to the complete model imported from Paper I. The calculation is elementary, but fixing every scale at this point is essential: the same constants enter the Poisson kernel, the conormal derivative, and the boundary energy in the next two sections.

Throughout this section and Sections 7– 8, assume

$$\mu > 0, \qquad \lambda > 0,$$

and let

$$\mathbb{H} = \{(x,y) \in \mathbb{R}^2 : y > 0\}, \qquad g_{\mu,\lambda} = \frac{1}{y^2}\left(\frac{dx^2}{\mu^2} + \frac{dy^2}{\lambda^2}\right), \qquad a(x,y) = -\frac{x}{y}.$$

The orientation is that of $dx \wedge dy$. We continue to use the divergence-of-gradient convention for Δ_g.

6.1 Conformal normalization and the arithmetic frame

Define the globally scaled coordinate and its complexification by

$$X = \frac{\lambda}{\mu} x, \qquad w = X + iy. \tag{100}$$

Proposition 6.1 (Analytic normalization of the basic model)**.** *In the coordinates* (100)*,*

$$g_{\mu,\lambda} = \frac{1}{\lambda^2}\frac{dX^2 + dy^2}{y^2}, \tag{101}$$

$$\mathrm{dvol}_g = \frac{dx\,dy}{\mu\lambda y^2} = \frac{dX\,dy}{\lambda^2 y^2}, \tag{102}$$

$$\Delta_g = \mu^2 y^2 \partial_x^2 + \lambda^2 y^2 \partial_y^2 = \lambda^2 y^2(\partial_X^2 + \partial_y^2). \tag{103}$$

For $F \in C^1(\mathbb{H};\mathbb{C})$, its Dirichlet energy, without a factor $1/2$, is

$$\begin{aligned}\mathcal{E}_g(F) := \int_{\mathbb{H}} |dF|_g^2\,\mathrm{dvol}_g &= \int_{\mathbb{H}} \left(\frac{\mu}{\lambda}|F_x|^2 + \frac{\lambda}{\mu}|F_y|^2\right) dx\,dy \\ &= \int_{\mathbb{H}} (|F_X|^2 + |F_y|^2)\,dX\,dy,\end{aligned} \tag{104}$$

whenever any one of these integrals is finite.

Proof. Since $dx = (\mu/\lambda)dX$, substitution into the metric gives (101). In the original coordinates,

$$(g_{ij}) = \begin{pmatrix} (\mu^2 y^2)^{-1} & 0 \\ 0 & (\lambda^2 y^2)^{-1} \end{pmatrix}, \qquad \sqrt{\det g} = \frac{1}{\mu\lambda y^2}.$$

This proves (102). Moreover,

$$\sqrt{\det g}\, g^{xx} = \frac{\mu}{\lambda}, \qquad \sqrt{\det g}\, g^{yy} = \frac{\lambda}{\mu}$$

are constant. The coordinate formula

$$\Delta_g F = \frac{1}{\sqrt{\det g}}\partial_i(\sqrt{\det g}\, g^{ij}\partial_j F)$$

therefore gives the first expression in (103); the identity $\partial_x = (\lambda/\mu)\partial_X$ gives the second. Finally, contracting dF with the inverse metric and then multiplying by $d\mathrm{vol}_g$ proves the first integral in (104). Changing from x to X proves the last one. This is the two-dimensional conformal invariance of Dirichlet energy written with all AEG scale factors visible. □

The positively oriented orthonormal arithmetic frame is

$$X_u = -\mu y \partial_x = -\lambda y \partial_X, \qquad X_v = -\lambda y \partial_y. \tag{105}$$

It satisfies

$$[X_u, X_v] = \lambda X_u. \tag{106}$$

Consequently, the general surface formula of Section 3 becomes

$$\Delta_g = X_u^2 + X_v^2 + \lambda X_v. \tag{107}$$

The drift λX_v cancels the first-order term created when X_v^2 is expanded in the y-coordinate. Thus the bare sum $X_u^2 + X_v^2$ is not the Laplace–Beltrami operator on this surface.

6.2 Complex structure and the assignment

Let

$$\partial_{\bar{w}} = \frac{1}{2}(\partial_X + i\partial_y), \qquad \partial_w = \frac{1}{2}(\partial_X - i\partial_y).$$

Equations (105) give

$$\overline{\partial}_{\mathrm{AEG}} = \frac{1}{2}(X_u + iX_v) = -\lambda y\, \partial_{\bar{w}}, \qquad \partial_{\mathrm{AEG}} = -\lambda y\, \partial_w. \tag{108}$$

Since λy never vanishes, arithmetic holomorphicity on the basic model is exactly ordinary holomorphicity in w. In particular, its real and imaginary parts are Δ_g-harmonic.

The assignment has the following two useful forms:

$$a = -\frac{\mu}{\lambda}\frac{X}{y}, \qquad w = y\left(i - \frac{\lambda}{\mu}a\right). \tag{109}$$

It is not itself harmonic. Directly from (103),

$$|\nabla a|_g^2 = \mu^2 + \lambda^2 a^2, \qquad \Delta_g a = 2\lambda^2 a. \tag{110}$$

The second identity is a pointwise eigen-equation in the sign convention of Paper I. It is not an L^2 spectral assertion: a is unbounded, has no finite continuous trace on the ideal boundary, and does not have finite global Dirichlet energy.

6.3 The ideal boundary

The conformal compactification is

$$\overline{\mathbb{H}}^{\mathrm{conf}} = \mathbb{H} \cup \mathbb{R} \cup \{\infty\},$$

where $\mathbb{R}$ is represented by $y = 0$ and the two ends of that line meet at the single point ∞. This is the closed disc after a Cayley transform. The line $y = 0$ is not a finite Riemannian boundary component: it lies at infinite $g_{\mu,\lambda}$-distance. Boundary conditions below are therefore conformal or ideal-boundary conditions.

The normalized coordinate $X = (\lambda/\mu)x$ extends to a homeomorphism of the compactified boundary and fixes the point ∞. Thus the classical upper-half-plane kernels may be transported to the original arithmetic x-coordinate, but the boundary density must be transformed as well. That transformation is carried out explicitly in the next section.

7 Poisson kernels and the ideal-boundary Dirichlet problem

Because the Laplace–Beltrami equation in (103) differs from the Euclidean Laplace equation only by the positive factor $\lambda^2 y^2$, the ideal-boundary Dirichlet problem is governed by the ordinary upper-half-plane Poisson kernel. The content of this section is the exact transport to the AEG normalization and a domain statement that includes the point at infinity.

7.1 The kernel in both boundary coordinates

For $X, t \in \mathbb{R}$ and $y > 0$, put

$$P_y(X-t) = \frac{1}{\pi}\frac{y}{(X-t)^2+y^2}. \tag{111}$$

For the original x-coordinate, define

$$P_y^{\mu,\lambda}(x-\xi) := \frac{1}{\pi}\frac{(\mu/\lambda)y}{(x-\xi)^2+(\mu y/\lambda)^2} = \frac{\mu\lambda y}{\pi(\lambda^2(x-\xi)^2+\mu^2y^2)}. \tag{112}$$

Lemma 7.1 (Kernel transport and normalization)**.** *If $X = (\lambda/\mu)x$ and $t = (\lambda/\mu)\xi$, then*

$$P_y(X-t)\,dt = P_y^{\mu,\lambda}(x-\xi)\,d\xi. \tag{113}$$

In particular,

$$P_y^{\mu,\lambda} \geq 0, \qquad \int_{\mathbb{R}} P_y^{\mu,\lambda}(x-\xi)\,d\xi = 1.$$

For fixed ξ, the function $(x,y) \mapsto P_y^{\mu,\lambda}(x-\xi)$ is Δ_g-harmonic on $\mathbb{H}$.

Proof. Substitute $X - t = (\lambda/\mu)(x-\xi)$ and $dt = (\lambda/\mu)d\xi$ into (111). This gives (113) and (112). Positivity is immediate, and normalization follows either from the same change of variables or from

$$\frac{1}{\pi}\int_{\mathbb{R}}\frac{h}{s^2+h^2}\,ds = 1 \qquad (h>0).$$

Finally, $P_y(X-t)$ is Euclidean harmonic away from its boundary pole. Equation (103) transports that assertion to $\Delta_g P_y^{\mu,\lambda} = 0$. □

7.2 The ideal-boundary Dirichlet theorem

We write $C_0(\mathbb{R})$ for the continuous functions satisfying $\varphi(\xi) \to 0$ as $|\xi| \to \infty$. Equivalently, such a function extends continuously to $\mathbb{R} \cup \{\infty\}$ with value zero at ∞.

Theorem 7.2 (C_0 ideal-boundary Dirichlet problem)**.** *For every $\varphi \in C_0(\mathbb{R})$, the function*

$$(\mathcal{P}_{\mu,\lambda}\varphi)(x,y) := \int_{\mathbb{R}} P_y^{\mu,\lambda}(x-\xi)\varphi(\xi)\,d\xi \tag{114}$$

has the following properties.

1. *It belongs to $C^\infty(\mathbb{H}) \cap C(\overline{\mathbb{H}}^{\mathrm{conf}})$ and is Δ_g-harmonic.*
2. *It takes the boundary values*

$$(\mathcal{P}_{\mu,\lambda}\varphi)(\xi,0) = \varphi(\xi) \quad (\xi \in \mathbb{R}), \qquad (\mathcal{P}_{\mu,\lambda}\varphi)(\infty) = 0.$$

3. *It satisfies the contraction estimate*

$$\|\mathcal{P}_{\mu,\lambda}\varphi\|_{L^\infty(\mathbb{H})} \leq \|\varphi\|_{L^\infty(\mathbb{R})}. \tag{115}$$

4. *It is the unique function in $C^2(\mathbb{H}) \cap C(\overline{\mathbb{H}}^{\mathrm{conf}})$ having these boundary values and satisfying $\Delta_g U = 0$.*

Proof. Put

$$\psi(t) = \varphi\left(\frac{\mu}{\lambda}t\right).$$

Then $\psi \in C_0(\mathbb{R})$, and Lemma 7.1 rewrites (114) as

$$U(X,y) = \int_{\mathbb{R}} P_y(X-t)\psi(t)\,dt. \tag{116}$$

For fixed $y > 0$, derivatives of the kernel are integrable after localization, and the usual differentiation argument shows that U is smooth and Euclidean harmonic. Equivalently, one may first prove this for compactly supported smooth data and use local convergence of the differentiated kernels. Equation (103) then gives $\Delta_g U = 0$. Positivity and unit mass of the kernel prove (115).

The kernels P_y form an approximate identity. Hence, at each $X_0 \in \mathbb{R}$,

$$U(X,y) - \psi(X_0) = \int_{\mathbb{R}} P_y(X-t)\big(\psi(t)-\psi(X_0)\big)\,dt \longrightarrow 0$$

as $(X,y) \to (X_0, 0)$. To see the behavior at infinity without assuming differentiability of the data, choose $\psi_0 \in C_c(\mathbb{R})$ with $\|\psi - \psi_0\|_\infty < \varepsilon$. The contraction estimate reduces the claim to ψ_0. If $\operatorname{supp}\psi_0 \subset [-R, R]$, then its Poisson integral tends to zero as $|X + iy| \to \infty$. Indeed, for large y it is bounded by $\|\psi_0\|_{L^1}/(\pi y)$; for large $|X|$, maximizing $y/((X-t)^2+y^2)$ over $y > 0$ gives a bound of order $(|X| - R)^{-1}$. Thus U extends continuously to the compactification with value zero at ∞.

For uniqueness, let V be the difference of two solutions. A Cayley map sends $\mathbb{H}$ to the unit disc and $\overline{\mathbb{H}}^{\mathrm{conf}}$ to the closed disc. The pullback of V is harmonic in the disc, continuous on its closure, and zero on the entire boundary circle. The maximum principle applied to its real and imaginary parts gives $V = 0$. □

Corollary 7.3 (Continuous data with a value at infinity)**.** *Let $\varphi \in C(\mathbb{R} \cup \{\infty\})$, so that*

$$L := \varphi(\infty) = \lim_{|\xi|\to\infty} \varphi(\xi).$$

Then

$$U = L + \mathcal{P}_{\mu,\lambda}(\varphi - L)$$

is the unique harmonic function continuous on $\overline{\mathbb{H}}^{\mathrm{conf}}$ with boundary value φ.

Proof. The function $\varphi - L$ belongs to $C_0(\mathbb{R})$, while constants are harmonic. Apply Theorem 7.2. □

Warning 7.4 (Why the function class matters). If neither boundedness nor a condition at the ideal point is imposed, uniqueness fails. The function $U(x,y) = y$ is nonzero and harmonic, extends continuously with value zero to every finite boundary point, and is unbounded at the ideal point. A separately prescribed bounded boundary value at the single point ∞ also cannot alter a bounded Poisson solution: that point has zero harmonic measure. Continuous data on the compactified boundary must have the same limit along both ends of the real line.

The theorem is deliberately a boundary theorem, not an L^2 spectral statement. Unless its boundary data vanish in a sufficiently strong sense, a Poisson extension is generally not in $L^2(\mathbb{H}, d\mathrm{vol}_g)$ because the hyperbolic volume density grows like y^{-2} near the ideal boundary.

8 Boundary energy and explicit arithmetic families

The Poisson theorem has two complementary refinements. Fourier analysis identifies its boundary energy operator exactly, while the assignment relation (109) produces explicit arithmetic families. We keep their domains separate: a plane wave is a useful bounded Fourier mode but does not extend continuously to the single ideal point ∞, whereas the C_0 theorem of the preceding section does.

8.1 Fourier representation and the conormal derivative

Use the Fourier convention

$$\widehat{\varphi}(\kappa) = \int_{\mathbb{R}} e^{-i\kappa x}\varphi(x)\,dx, \qquad \varphi(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{i\kappa x}\widehat{\varphi}(\kappa)\,d\kappa. \tag{117}$$

Write $|D_x|$ for the multiplier with symbol $|\kappa|$.

Proposition 8.1 (Fourier formula for the Poisson extension)**.** *If $\varphi \in \mathcal{S}(\mathbb{R})$ and $U = \mathcal{P}_{\mu,\lambda}\varphi$, then*

$$U(x,y) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{\varphi}(\kappa)e^{i\kappa x-(\mu/\lambda)|\kappa|y}\,d\kappa. \tag{118}$$

Consequently,

$$-\partial_y U(\,\cdot\,,0) = \frac{\mu}{\lambda}|D_x|\varphi \tag{119}$$

in $\mathcal{S}(\mathbb{R})$ and in $L^2(\mathbb{R})$.

Proof. The Fourier transform in x of $\pi^{-1}h/(x^2+h^2)$ is $e^{-h|\kappa|}$. In (112), $h = (\mu/\lambda)y$. The convolution theorem therefore gives (118). Since the data are Schwartz, one may differentiate under the integral for $y \geq 0$ in L^2; evaluation at $y = 0$ yields (119). □

The raw derivative in (119) is not yet the boundary operator paired with the Riemannian energy. On the truncated domain $\{y \geq \varepsilon\}$, the outward g-unit normal and induced measure along $y = \varepsilon$ are

$$\nu = -\lambda y\partial_y, \qquad d\sigma_g = \frac{dx}{\mu y}. \tag{120}$$

Thus

$$(\partial_\nu U)d\sigma_g = -\frac{\lambda}{\mu}U_y\,dx.$$

This fixes the conformal conormal scale.

Definition 8.2 (Conformal Dirichlet-to-Neumann operator)**.** For $\varphi \in \mathcal{S}(\mathbb{R})$, define

$$\mathcal{N}_x\varphi := -\frac{\lambda}{\mu}\partial_y(\mathcal{P}_{\mu,\lambda}\varphi)(\,\cdot\,,0). \tag{121}$$

Theorem 8.3 (Dirichlet-to-Neumann and energy identity)**.** *For every $\varphi \in \mathcal{S}(\mathbb{R})$,*

$$\mathcal{N}_x\varphi = |D_x|\varphi, \tag{122}$$

$$\mathcal{E}_g(\mathcal{P}_{\mu,\lambda}\varphi) = \int_{\mathbb{R}} \overline{\varphi(x)}\,\mathcal{N}_x\varphi(x)\,dx = \frac{1}{2\pi}\int_{\mathbb{R}} |\kappa|\,|\widehat{\varphi}(\kappa)|^2\,d\kappa. \tag{123}$$

For real-valued data the complex conjugate may be omitted. In the normalized X-coordinate, the same statement reads

$$\mathcal{N}_X = -\partial_y\big|_{y=0} = |D_X|.$$

Proof. Equation (122) follows immediately from (119) and (121). To calculate the energy, insert (118) into (104) and use Plancherel. With $c = \mu/\lambda$,

$$\begin{aligned}\mathcal{E}_g(U) &= \frac{1}{2\pi}\int_{\mathbb{R}}\int_0^\infty \left(c\kappa^2 + c^{-1}c^2\kappa^2\right)|\widehat{\varphi}(\kappa)|^2 e^{-2c|\kappa|y}\,dy\,d\kappa \\ &= \frac{1}{2\pi}\int_{\mathbb{R}} |\kappa||\widehat{\varphi}(\kappa)|^2\,d\kappa.\end{aligned}$$

Plancherel once more identifies this expression with $\int \overline{\varphi}\,|D_x|\varphi\,dx$. Finally, if $\psi(X) = \varphi((\mu/\lambda)X)$, the X-Fourier solution decays as $e^{-|\xi|y}$, so its raw outward conformal derivative is $|D_X|$. ☐

The right side of (123) is the homogeneous $H^{1/2}(\mathbb{R})$ quadratic form. The identity therefore extends by completion from Schwartz data to boundary data with finite homogeneous $H^{1/2}$ seminorm, modulo the usual treatment of constants.

Theorem 8.4 (Energy-minimizing extension)**.** *Let $\varphi \in \mathcal{S}(\mathbb{R})$ and $U = \mathcal{P}_{\mu,\lambda}\varphi$. For every $h \in H^1_{0,\mathrm{Euc}}(\mathbb{H})$, where this denotes the Euclidean half-plane Sobolev space, the function $U + h$ has the same trace as U and*

$$\mathcal{E}_g(U + h) = \mathcal{E}_g(U) + \mathcal{E}_g(h). \tag{124}$$

Consequently, U is the unique energy minimizer in the affine class

$$U + H^1_{0,\mathrm{Euc}}(\mathbb{H}).$$

It is also the unique weakly Δ_g-harmonic member of that class.

Proof. The anisotropic energy norm in (104) is equivalent to the Euclidean gradient norm because $\mu, \lambda > 0$. For $h \in C_c^\infty(\mathbb{H})$, integration by parts and $\Delta_g U = 0$ give

$$\int_{\mathbb{H}} \left(\frac{\mu}{\lambda}U_x\overline{h_x} + \frac{\lambda}{\mu}U_y\overline{h_y}\right)dx\,dy = 0.$$

Density of $C_c^\infty(\mathbb{H})$ in $H^1_{0,\mathrm{Euc}}(\mathbb{H})$ extends this orthogonality to all admissible h. Expanding the squared gradient proves (124). Equality of energies forces $\nabla h = 0$ almost everywhere; since $h \in H^1_{0,\mathrm{Euc}}(\mathbb{H}) \subset L^2(\mathbb{H})$, the constant is zero. This proves uniqueness of the minimizer.

If $U + h$ is also weakly harmonic, subtract the weak equations for $U + h$ and U, and test the resulting equation for h by h itself, using density. Then $\mathcal{E}_g(h) = 0$, so $h = 0$. ☐

8.2 Holomorphic and Fourier families in assignment coordinates

Equation (108) immediately gives a broad family, with an explicit arithmetic presentation.

Proposition 8.5 (Assignment presentation of holomorphic fields)**.** *Let $\Omega \subset \mathbb{C}$ be open and let Φ be holomorphic on Ω. On every component of*

$$\left\{(a, y) \in \mathbb{R} \times (0, \infty) : y\left(i - \frac{\lambda}{\mu}a\right) \in \Omega\right\},$$

the field

$$F(a, y) = \Phi\left(y\left(i - \frac{\lambda}{\mu}a\right)\right) \tag{125}$$

is arithmetic holomorphic. Its real and imaginary parts are Δ_g-harmonic.

Proof. The argument of Φ is precisely w by (109). Hence $\partial_{\bar{w}}\Phi(w) = 0$, and (108) gives $\overline{\partial}_{\mathrm{AEG}}F = 0$. Surface holomorphicity implies Laplace–Beltrami harmonicity by Section 3. ☐

Examples include polynomials in w and rational functions $R(w)$ on domains avoiding their poles. If all poles of R lie in $\mathbb{C} \setminus \overline{\mathbb{H}}$, then $R(w)$ is globally holomorphic on $\mathbb{H}$. For example, for every $\tau > 0$,

$$R_\tau(w) = \frac{1}{w + i\tau} = \frac{1}{-(\lambda/\mu)ay + i(y+\tau)} \tag{126}$$

is a global arithmetic holomorphic field; its pole $-i\tau$ lies in the lower half-plane. Its real and imaginary parts are explicit rational harmonic functions of (a, y). The positive-frequency modes are, for $\kappa > 0$,

$$H_\kappa(x, y) = e^{i\kappa x - (\mu/\lambda)\kappa y} = e^{i(\mu/\lambda)\kappa w} = e^{-i\kappa a y - (\mu/\lambda)\kappa y}. \tag{127}$$

They are holomorphic. The decaying modes with negative frequency are anti-holomorphic. Formula (118) is their superposition. Individual plane waves have bounded boundary values but no limit at the compactification point ∞, so they are not examples in the C_0 theorem.

8.3 Assignment-only harmonic fields

The assignment itself is not harmonic, but one nonlinear rectification of it is. In fact there are no others beyond affine recombination.

Theorem 8.6 (Assignment-only harmonic classification)**.** *Let $I \subset \mathbb{R}$ be an open interval, let $h \in C^2(I)$, and put $\Omega_I = a^{-1}(I)$. Then*

$$\Delta_g(h \circ a) = (\mu^2 + \lambda^2 a^2)h''(a) + 2\lambda^2 a h'(a). \tag{128}$$

Consequently,

$$\Delta_g(h \circ a) = 0 \quad \Longleftrightarrow \quad h(a) = C_0 + C_1 \arctan\left(\frac{\lambda a}{\mu}\right) \tag{129}$$

for constants $C_0, C_1 \in \mathbb{C}$.

Proof. The chain rule for the Laplace–Beltrami operator and (110) give

$$\Delta_g(h \circ a) = h''(a)|\nabla a|_g^2 + h'(a)\Delta_g a,$$

which is (128). If it vanishes and $p = h'$, then

$$(\mu^2 + \lambda^2 a^2)p'(a) + 2\lambda^2 a p(a) = 0.$$

Thus

$$p(a) = \frac{C}{\mu^2 + \lambda^2 a^2},$$

and integration yields

$$h(a) = C_0 + \frac{C}{\mu\lambda}\arctan\left(\frac{\lambda a}{\mu}\right).$$

Absorbing the nonzero constant $(\mu\lambda)^{-1}$ into C_1 proves the classification on I. Here no values of h are left untested: the assignment $a = -x/y$ maps Ω_I onto I. The converse follows by differentiation. ☐

Corollary 8.7 (A half-line harmonic measure)**.** *The bounded function*

$$\omega_-(a) = \frac{1}{2} + \frac{1}{\pi}\arctan\left(\frac{\lambda a}{\mu}\right) \tag{130}$$

is the Poisson extension of the almost-everywhere boundary datum $\mathbf{1}_{(-\infty,0)}$ *in the* x*-coordinate. Equivalently,*

$$\omega_-(a) = \frac{\operatorname{Arg} w}{\pi}, \qquad 0 < \operatorname{Arg} w < \pi. \tag{131}$$

Proof. Writing $w = re^{i\theta}$ with $0 < \theta < \pi$, one has

$$\frac{\lambda a}{\mu} = -\frac{X}{y} = -\cot\theta.$$

On this range, $\arctan(-\cot\theta) = \theta - \pi/2$, which proves (131). The angle θ/π tends to zero on the positive half-line and to one on the negative half-line. It is bounded and harmonic. Directly, with $c = \mu/\lambda$,

$$\begin{aligned}\int_{-\infty}^{0} P_y^{\mu,\lambda}(x-\xi)\,d\xi &= \frac{1}{\pi}\left[\arctan\left(\frac{\xi - x}{cy}\right)\right]_{\xi=-\infty}^{\xi=0} \\ &= \frac{1}{2} - \frac{1}{\pi}\arctan\left(\frac{x}{cy}\right) = \frac{1}{2} + \frac{1}{\pi}\arctan\left(\frac{\lambda a}{\mu}\right).\end{aligned}$$

This identifies it as the stated harmonic measure without imposing continuity at the jump point. □

The holomorphic function behind this corollary is the principal logarithm on the upper half-plane:

$$\operatorname{Log} w = \log y + \frac{1}{2}\log\left(1 + \frac{\lambda^2 a^2}{\mu^2}\right) + i\left[\frac{\pi}{2} + \arctan\left(\frac{\lambda a}{\mu}\right)\right]. \tag{132}$$

The example lies outside the continuous compactified-boundary and finite-energy classes. Its boundary datum jumps at $x = 0$ and has incompatible limits along the two ends that meet at ∞. Moreover, in polar coordinates for w,

$$\int_{\mathbb{H}} |\nabla(\operatorname{Arg} w)|^2\,dX\,dy = \int_0^{\pi}\int_0^{\infty} \frac{1}{r^2} r\,dr\,d\theta = \infty.$$

Thus the half-line harmonic measure is a valid bounded Poisson solution for L^∞ data, but not a member of the energy theorem above. These domain distinctions also explain why the unbounded assignment $a = -x/y$ itself cannot serve as ideal-boundary Dirichlet data.

9 Conclusions and analytic frontier

The analytic question posed at the beginning of this paper has two answers, because Paper I supplies two geometrically different carriers for analysis. A regular AES is an oriented Riemannian surface whose metric, orientation, assignment, and arithmetic frame are already fixed. Its complex analysis is elliptic and intrinsic. The contact state space is three-dimensional and nonintegrable; its CR and sub-Riemannian analysis starts only after a horizontal metric and a measure have been declared. Keeping these carriers separate is the main organizing decision of the paper.

On the surface branch, the arithmetic frame makes the first-order drift of the Laplace–Beltrami operator visible. If $[X_u, X_v] = c_u X_u + c_v X_v$, then

$$\Delta_g = X_u^2 + X_v^2 - c_v X_u + c_u X_v.$$

This formula resolves the ambiguity between a raw sum of squares and the divergence-form operator selected by Riemannian volume. The same structure coefficients give an exact Cauchy–Riemann factorization, and hence a direct proof that arithmetic holomorphic fields are Laplace–Beltrami harmonic. The unitary gauge relation with the ordinary Riemann-surface $\overline{\partial}$-operator also fixes the strength of this statement: arithmetic holomorphicity is not a new holomorphic function class. It is the classical class expressed in the canonical frame selected by the assignment law.

On the contact branch, contact volume selects

$$\Delta_{\mathcal{C}} = D_u^2 + D_v^2 + \lambda D_v,$$

whereas the raw sum $D_u^2 + D_v^2$ is the expression appearing in the elementary CR product. The two roles are compatible but not interchangeable. The adjoint factorization displays the Reeb correction, while the raw factorization displays the vertical curvature $\mu\lambda\partial_a$. The logarithmic first integrals and affine–Appell modules are consequently explicit CR constructions on the contact space, not harmonic functions on a hidden horizontal surface.

9.1 What is established on the basic model

The complete basic hyperbolic AES provides the global test case. After the fixed normalization $X = (\lambda/\mu)x$, it is a constant curvature rescaling of the standard upper half-plane. This observation is elementary, but its consequences for the AEG variables are precise. The arithmetic coordinate is

$$w = X + iy = y\left(i - \frac{\lambda}{\mu}a\right),$$

the ideal-boundary Poisson kernel solves the C_0 Dirichlet problem uniquely, and the variational conormal derivative has symbol $|D|$. For Poisson extensions, the exact energy identity is precisely the homogeneous $H^{1/2}$ boundary quadratic form and extends by completion from Schwartz data. Thus the basic model supports more than a formal differential calculus: it has a proved boundary kernel, a uniqueness class, a Fourier-multiplier Dirichlet-to-Neumann map on Schwartz data, and a variational interpretation.

The assignment does not merely relabel these results. In the (a, y) chart it turns every admissible holomorphic function of w into an explicitly assignment-dependent field. Positive-frequency modes and rational fields supply concrete families, while the ordinary differential equation for $h(a)$ gives the complete classification

$$h(a) = A + B\arctan\left(\frac{\lambda a}{\mu}\right)$$

of harmonic fields depending only on the assignment. The nonconstant member is a boundary half-line harmonic measure. Its discontinuous trace also shows why the function class must be stated in every boundary theorem.

9.2 Holomorphic pullback as an analytic export

The planar and cylindrical constructions add a second kind of conclusion to the global boundary theorem. They show that regular AES data can be transported functorially by a locally biholomorphic

Feature	Regular-AES surface	Contact state space
Carrier	oriented Riemannian surface (M, g)	contact three-manifold $(\mathcal{C}, \ker \alpha)$
Preferred fields	tangent frame (X_u, X_v)	horizontal lifts (D_u, D_v)
Complex datum	determined by g and orientation	chosen from the normalized horizontal metric and orientation
Natural measure	Riemannian area $d\mathrm{vol}_g$	positive contact volume $\lvert\alpha \wedge d\alpha\rvert$
Variational operator	Δ_g, including structure drift	$\Delta_{\mathcal{C}} = D_u^2 + D_v^2 + \lambda D_v$
First-order kernel	ordinary holomorphic class in arithmetic gauge	contact-CR class with Reeb/vertical twist
Global result here	Poisson, Dirichlet, energy, and DtN on the basic model	Friedrichs realization, hypoellipticity, and explicit CR fields

Table 1: Analytic data and conclusions on the two carriers. Similar first-order notation does not identify the two geometries.

map. For the planar harmonic base, the pullback preserves

$$\Delta_g a = 0, \qquad K_g = \lambda^2 \frac{\mu^2 - \lambda^2 a^2}{\mu^2 + \lambda^2 a^2}.$$

For the complete flat cylinder, it preserves

$$\Delta_g a = \lambda^2 a, \qquad Z(a) = W^{-1}(S^1).$$

These formulas give an analytic interface to automorphic constructions: unitary multipliers preserve the full AES data, while inversion preserves the metric and zero set and reverses the assignment. The latter case is naturally expressed by a sign line bundle on the quotient rather than by silently forcing a scalar-valued assignment.

This export has a strict boundary. At a critical point of the holomorphic map the pullback metric degenerates. At a zero or pole of the cylindrical target map the logarithmic assignment ceases to be finite, and an algebraic branch point requires a choice of local sheet. Paper II proves no continuation through these sets. It supplies the exact formulas on their complement. Paper III uses this interface to analyze local branched cones and the specific $q = 4$ sign cover; its successful completion in that model is not a general continuation theorem for other branch sets.

Table 1 summarizes the two analytic branches and the operator choices proved in this paper.

9.3 Analytic frontier

The results above close the first function-theoretic layer, but they do not close the programme. The following problems remain genuinely open within the scope of Paper II.

Open problem 9.1 (General boundary theory)*.* Identify geometric hypotheses on a complete regular AES under which a meaningful ideal boundary, harmonic measure, Green kernel, and a well-posed Dirichlet class can be constructed. The assignment law alone does not provide such a compactification.

Open problem 9.2 (Spectral realization beyond the basic model)*.* Determine the self-adjoint spectral theory of $-\Delta_g$ and of the contact Friedrichs operator for geometrically controlled nonhomogeneous examples. In particular, separate pointwise eigen-equations from L^2 eigenvectors and specify the boundary conditions of every realization.

Open problem 9.3 (Contact boundary and completeness). Develop a boundary-value theory for the normalized contact-CR structure and decide whether an appropriate completion of the explicit logarithmic and Fourier–CR families gives a useful representation theorem. No completeness assertion is made here.

Open problem 9.4 (Degenerate and singular assignments). Extend the analytic definitions across points where the regular frame ceases to exist. This requires the singular geometric input assigned to Paper III; the present paper neither assumes nor proves a removable-singularity theorem there.

Open problem 9.5 (Automorphic singular completion). Let an automorphic or algebraic function produce a cylindrical pullback on the regular locus defined by $0 < |W| < \infty$ and $dW \neq 0$. Determine local and global conditions under which the metric, assignment, or assignment line bundle admits a singular AES completion. Relate ramification indices to zero-graph valence without assuming that every critical point lies over the unit circle.

The dependency boundary is therefore explicit. Paper II uses only the regular surface, hyperbolic model, and contact connection proved in Paper I. It exports elliptic and contact analytic tools, together with planar and cylindrical pullback formulas, that may later be tested on singular families. It does not import tube topology from Paper III or projective-complexity claims from Paper IV. Within that boundary, every advertised result is backed by a stated measure, regular locus, the relevant test-function or variational domain, and a proof. This is the appropriate starting point for a broader arithmetic real function theory.

A Frame, adjoint, and factorization calculations

This appendix gives the sign-sensitive calculations behind Propositions 2.1 and 2.2 and Theorem 3.6. All calculations are local, but the resulting identities are intrinsic.

A.1 Connection coefficients from the frame bracket

Let $e_1 = X_u$, $e_2 = X_v$, and write

$$[e_1, e_2] = c_u e_1 + c_v e_2. \tag{133}$$

Let ω be the Levi–Civita connection one-form in the convention

$$\nabla_Y e_1 = \omega(Y) e_2, \qquad \nabla_Y e_2 = -\omega(Y) e_1. \tag{134}$$

Torsion freeness gives

$$\begin{aligned} [e_1, e_2] &= \nabla_{e_1} e_2 - \nabla_{e_2} e_1 \\ &= -\omega(e_1) e_1 - \omega(e_2) e_2. \end{aligned}$$

Comparison with (133) yields

$$\omega(e_1) = -c_u, \qquad \omega(e_2) = -c_v. \tag{135}$$

Therefore

$$\nabla_{e_1} e_1 = -c_u e_2, \qquad \nabla_{e_1} e_2 = c_u e_1, \tag{136}$$

$$\nabla_{e_2} e_1 = -c_v e_2, \qquad \nabla_{e_2} e_2 = c_v e_1. \tag{137}$$

Taking traces gives

$$\begin{aligned} \operatorname{div}_g e_1 &= \langle \nabla_{e_1} e_1, e_1 \rangle + \langle \nabla_{e_2} e_1, e_2 \rangle = -c_v, \\ \operatorname{div}_g e_2 &= \langle \nabla_{e_1} e_2, e_1 \rangle + \langle \nabla_{e_2} e_2, e_2 \rangle = c_u. \end{aligned}$$

This proves the divergence signs in (9) without a coordinate choice.

A.2 Integration by parts and the real energy operator

For a real vector field X and compactly supported complex-valued functions f, h, apply the divergence theorem to $f\overline{h}X$:

$$\begin{aligned} 0 &= \int_M \operatorname{div}_g(f\overline{h}X)\, d\mathrm{vol}_g \\ &= \int_M \big((Xf)\overline{h} + f\, X(\overline{h}) + f\overline{h}\, \operatorname{div}_g X\big)\, d\mathrm{vol}_g. \end{aligned}$$

Since $X(\overline{h}) = \overline{Xh}$,

$$\int_M (Xf)\overline{h}\, d\mathrm{vol}_g = \int_M f\, \overline{(-X - \operatorname{div}_g X)h}\, d\mathrm{vol}_g. \tag{138}$$

Thus $X^* = -X - \operatorname{div}_g X$. Inserting the two divergences above gives

$$e_1^* = -e_1 + c_v, \qquad e_2^* = -e_2 - c_u.$$

Consequently,

$$\begin{aligned} e_1^* e_1 + e_2^* e_2 &= -e_1^2 - e_2^2 + c_v e_1 - c_u e_2 \\ &= -(e_1^2 + e_2^2 - c_v e_1 + c_u e_2). \end{aligned}$$

On the other hand,

$$\begin{aligned} \Delta_g f &= \operatorname{div}_g\big((e_1 f)e_1 + (e_2 f)e_2\big) \\ &= e_1^2 f + e_2^2 f + (\operatorname{div}_g e_1)e_1 f + (\operatorname{div}_g e_2)e_2 f \\ &= e_1^2 f + e_2^2 f - c_v e_1 f + c_u e_2 f. \end{aligned}$$

This verifies both (12) and (13).

A.3 Raw complex products

Put

$$A = e_1 - ie_2 = 2\partial_{\mathrm{AEG}}, \qquad B = e_1 + ie_2 = 2\overline{\partial}_{\mathrm{AEG}}.$$

Then

$$\begin{aligned} AB &= (e_1 - ie_2)(e_1 + ie_2) \\ &= e_1^2 + e_2^2 + i(e_1 e_2 - e_2 e_1) \\ &= S_M + i(c_u e_1 + c_v e_2), \end{aligned} \tag{139}$$

whereas

$$BA = S_M - i(c_u e_1 + c_v e_2). \tag{140}$$

These formulas use only the differential expressions; neither a volume measure nor a domain has entered.

A.4 Complex formal adjoints and cancellation of the drift

With the complex L^2 inner product, scalar coefficients are conjugated when an adjoint is taken. Therefore

$$\begin{aligned}\overline{\partial}^*_{\mathrm{AEG}} &= \frac{1}{2}(e_1^* - ie_2^*)\\ &= \frac{1}{2}(-e_1 + c_v + ie_2 + ic_u)\\ &= -\partial_{\mathrm{AEG}} + \frac{1}{2}(c_v + ic_u),\end{aligned}$$

and

$$\begin{aligned}\partial^*_{\mathrm{AEG}} &= \frac{1}{2}(e_1^* + ie_2^*)\\ &= -\overline{\partial}_{\mathrm{AEG}} + \frac{1}{2}(c_v - ic_u).\end{aligned}$$

For the first quadratic product, use (139):

$$\begin{aligned}4\overline{\partial}^*_{\mathrm{AEG}}\overline{\partial}_{\mathrm{AEG}} &= (-A + c_v + ic_u)B\\ &= -AB + (c_v + ic_u)(e_1 + ie_2)\\ &= -S_M - i(c_u e_1 + c_v e_2)\\ &\quad + c_v e_1 + ic_v e_2 + ic_u e_1 - c_u e_2\\ &= -S_M + c_v e_1 - c_u e_2\\ &= -\Delta_g.\end{aligned}$$

For the conjugate product, use (140):

$$\begin{aligned}4\partial^*_{\mathrm{AEG}}\partial_{\mathrm{AEG}} &= (-B + c_v - ic_u)A\\ &= -BA + (c_v - ic_u)(e_1 - ie_2)\\ &= -S_M + i(c_u e_1 + c_v e_2)\\ &\quad + c_v e_1 - ic_v e_2 - ic_u e_1 - c_u e_2\\ &= -S_M + c_v e_1 - c_u e_2\\ &= -\Delta_g.\end{aligned}$$

This term-by-term cancellation is the reason the raw bracket twist and the Riemannian drift must not be treated as competing definitions of a Laplacian.

A.5 Gauge check

Let

$$e_1' = \cos\vartheta\, e_1 + \sin\vartheta\, e_2, \qquad e_2' = -\sin\vartheta\, e_1 + \cos\vartheta\, e_2.$$

Then

$$\begin{aligned}e_1' + ie_2' &= (\cos\vartheta - i\sin\vartheta)e_1 + (\sin\vartheta + i\cos\vartheta)e_2\\ &= e^{-i\vartheta}(e_1 + ie_2).\end{aligned}$$

If (η^1, η^2) and $(\eta^{1'}, \eta^{2'})$ are the corresponding dual coframes, direct inversion of the rotation matrix gives

$$\eta^{1'} - i\eta^{2'} = e^{i\vartheta}(\eta^1 - i\eta^2).$$

Hence

$$\left(\frac{1}{2}(e_1' + ie_2')F\right)(\eta^{1'} - i\eta^{2'}) = \left(\frac{1}{2}(e_1 + ie_2)F\right)(\eta^1 - i\eta^2),$$

which is the explicit $U(1)$-gauge invariance of the intrinsic $\bar{\partial}_M F$.

A.6 Hyperbolic sign check

On the basic hyperbolic AES,

$$e_1 = -\mu y\partial_x, \qquad e_2 = -\lambda y\partial_y, \qquad [e_1, e_2] = \lambda e_1.$$

Thus $c_u = \lambda$ and $c_v = 0$. The general formulas reduce to

$$e_1^* = -e_1, \qquad\qquad e_2^* = -e_2 - \lambda,$$
$$\Delta_g = e_1^2 + e_2^2 + \lambda e_2, \qquad\qquad \overline{\partial}^*_{\mathrm{AEG}} = -\partial_{\mathrm{AEG}} + \frac{i\lambda}{2}.$$

Since

$$e_1^2 = \mu^2 y^2 \partial_x^2, \qquad e_2^2 = \lambda^2 y^2 \partial_y^2 + \lambda^2 y \partial_y, \qquad \lambda e_2 = -\lambda^2 y \partial_y,$$

the two first-order y-terms cancel and

$$\Delta_g = \mu^2 y^2 \partial_x^2 + \lambda^2 y^2 \partial_y^2,$$

in agreement with the coordinate calculation in Paper I.

B Contact-CR and affine–Appell calculations

This appendix records the coordinate calculations behind Section 4. They concern only the three-dimensional contact state space $\mathcal{C}$; none of the fields below is a surface frame field X_u or X_v.

B.1 Contact volume, brackets, and divergence

From

$$\alpha = da - \mu\, du - \lambda a\, dv, \qquad d\alpha = -\lambda\, da \wedge dv,$$

one obtains

$$\alpha \wedge d\alpha = -\mu\lambda\, du \wedge dv \wedge da. \tag{141}$$

Thus α is a contact form precisely when $\mu\lambda \neq 0$, and its positive contact density is $d\nu_{\mathrm{c}} = |\mu\lambda|\, du\, dv\, da$.

For completeness, the bracket calculation is

$$\begin{aligned}[D_u, D_v] &= [\partial_u + \mu\partial_a, \partial_v + \lambda a\partial_a] \\ &= \mu\, \partial_a(\lambda a)\partial_a = \mu\lambda T.\end{aligned} \tag{142}$$

In particular, $D_u, D_v, [D_u, D_v]$ have coefficient matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \mu & \lambda a & \mu\lambda \end{pmatrix}$$

relative to $\partial_u, \partial_v, \partial_a$, and its determinant is $\mu\lambda$. This also verifies the step-two bracket condition directly.

Because $d\nu_c$ has constant coordinate density,

$$\operatorname{div}_{\nu_c} D_u = \partial_u(1) + \partial_a(\mu) = 0, \qquad \operatorname{div}_{\nu_c} D_v = \partial_v(1) + \partial_a(\lambda a) = \lambda. \tag{143}$$

For $f, g \in C_c^\infty(\mathcal{C})$ and a real vector field Y, integration of $\operatorname{div}(f\overline{g}Y)$ gives

$$\int (Yf)\overline{g}\, d\nu_c = \int f\, \overline{(-Y - \operatorname{div}_{\nu_c} Y)g}\, d\nu_c. \tag{144}$$

Equations (143) and (144) prove the adjoint formulas in Proposition 4.3.

B.2 Expansion of the variational operator

The two squares expand as

$$\begin{aligned} D_u^2 &= \partial_u^2 + 2\mu\partial_u\partial_a + \mu^2\partial_a^2, \\ D_v^2 &= \partial_v^2 + 2\lambda a\partial_v\partial_a + \lambda^2a^2\partial_a^2 + \lambda^2 a\partial_a. \end{aligned}$$

Consequently the raw sum of squares is

$$Q_{\mathcal{C}} = \partial_u^2 + \partial_v^2 + 2\mu\partial_u\partial_a + 2\lambda a\partial_v\partial_a + (\mu^2 + \lambda^2a^2)\partial_a^2 + \lambda^2 a\partial_a, \tag{145}$$

whereas the divergence-of-gradient expression is

$$\Delta_{\mathcal{C}} = Q_{\mathcal{C}} + \lambda\partial_v + \lambda^2 a\partial_a. \tag{146}$$

The last two terms are the contact-volume drift. In particular, $Q_{\mathcal{C}}$ and $\Delta_{\mathcal{C}}$ are distinct when $\lambda \neq 0$.

For compactly supported f,

$$\begin{aligned} \langle -\Delta_{\mathcal{C}} f, f\rangle_{\nu_c} &= \langle D_u^* D_u f + D_v^* D_v f, f\rangle_{\nu_c} \\ &= \|D_u f\|^2_{L^2(\nu_c)} + \|D_v f\|^2_{L^2(\nu_c)}. \end{aligned} \tag{147}$$

This calculation identifies the differential expression represented by the closed energy form; it does not by itself assert essential self-adjointness of the minimal differential operator.

B.3 Raw and adjoint CR products

Writing $\overline{L} = \frac{1}{2}(D_u + iD_v)$ and $L = \frac{1}{2}(D_u - iD_v)$, direct multiplication gives

$$4L\overline{L} = (D_u - iD_v)(D_u + iD_v) = Q_{\mathcal{C}} + i[D_u, D_v] = Q_{\mathcal{C}} + i\mu\lambda T, \tag{148}$$

$$4\overline{L}L = (D_u + iD_v)(D_u - iD_v) = Q_{\mathcal{C}} - i[D_u, D_v] = Q_{\mathcal{C}} - i\mu\lambda T. \tag{149}$$

These are algebraic identities and do not depend on a measure.

The contact-volume adjoints do depend on the measure. Conjugating scalar coefficients and using $D_u^* = -D_u$ and $D_v^* = -D_v - \lambda$ yields

$$\overline{L}^* = \frac{1}{2}(-D_u + iD_v + i\lambda), \qquad L^* = \frac{1}{2}(-D_u - iD_v - i\lambda).$$

The first energy product is

$$\begin{aligned}4\overline{L}^*\overline{L} &= (-D_u + iD_v + i\lambda)(D_u + iD_v) \\ &= -D_u^2 - D_v^2 - \lambda D_v + i\lambda D_u - i[D_u, D_v] \\ &= A_{\mathcal{C}} + i\lambda D_u - i\mu\lambda T. \end{aligned} \tag{150}$$

Similarly,

$$\begin{aligned}4L^*L &= (-D_u - iD_v - i\lambda)(D_u - iD_v) \\ &= -D_u^2 - D_v^2 - \lambda D_v - i\lambda D_u + i[D_u, D_v] \\ &= A_{\mathcal{C}} - i\lambda D_u + i\mu\lambda T. \end{aligned} \tag{151}$$

Since $\mathcal{R} = -\mu^{-1}\partial_u$ and $D_u = \partial_u + \mu T$, these formulas are equivalently

$$4\overline{L}^*\overline{L} = A_{\mathcal{C}} - i\mu\lambda\mathcal{R}, \qquad 4L^*L = A_{\mathcal{C}} + i\mu\lambda\mathcal{R}. \tag{152}$$

The difference between Equations (148)–(149) and Equations (150)–(151) is therefore not a sign convention: the latter products incorporate the contact-volume adjoints.

B.4 The balanced-measure conjugation

Let $d\nu_{\mathrm{b}} = e^{-\lambda v} d\nu_{\mathrm{c}}$. For a weighted density $e^{\phi} d\nu_{\mathrm{c}}$,

$$\operatorname{div}_{e^{\phi}\nu_{\mathrm{c}}} Y = \operatorname{div}_{\nu_{\mathrm{c}}} Y + Y\phi. \tag{153}$$

Taking $\phi = -\lambda v$ gives $D_u\phi = 0$ and $D_v\phi = -\lambda$, so both horizontal divergences vanish for $d\nu_{\mathrm{b}}$. Hence the balanced-measure energy operator is the formal expression $-Q_{\mathcal{C}}$.

The unitary map

$$U : L^2(d\nu_{\mathrm{b}}) \longrightarrow L^2(d\nu_{\mathrm{c}}), \qquad Uf = e^{-\lambda v/2} f,$$

satisfies, on test functions,

$$UD_uU^{-1}f = D_uf, \tag{154}$$

$$UD_vU^{-1}f = D_vf + \frac{\lambda}{2}f. \tag{155}$$

It follows that

$$\begin{aligned}U(-Q_{\mathcal{C}})U^{-1} &= -D_u^2 - \left(D_v + \frac{\lambda}{2}\right)^2 \\ &= -D_u^2 - D_v^2 - \lambda D_v - \frac{\lambda^2}{4} \\ &= A_{\mathcal{C}} - \frac{\lambda^2}{4}. \end{aligned} \tag{156}$$

Thus changing the measure and applying its unitary gauge produces the constant zeroth-order shift in Equation (68); it does not identify the two unitarily conjugate differential expressions without that shift.

B.5 Logarithmic first integrals

Put $W(a) = \mu + i\lambda a$. Since $\mu \neq 0$, the image of W is contained in the half-plane Ω_μ used in Proposition 4.14; therefore the chosen branch $\mathrm{Log}_\mu W$ is smooth for every real a. For

$$z = u + iv, \qquad \zeta = u + \frac{i}{\lambda}\mathrm{Log}_\mu W(a),$$

one has

$$\begin{aligned} &D_u z = 1, && D_v z = i, && && (157)\\ &\partial_a \zeta = -\frac{1}{W}, && D_u\zeta = 1 - \frac{\mu}{W} = \frac{i\lambda a}{W}, && D_v\zeta = -\frac{\lambda a}{W}. && (158)\end{aligned}$$

Hence $D_u z + iD_v z = 0$ and $D_u\zeta + iD_v\zeta = 0$. This proves the CR assertions without treating ζ as a real coordinate on $\mathcal{C}$. Indeed, the useful global object is the rank-three map $(z, \zeta) : \mathcal{C} \to \mathbb{C}^2$.

For $\rho, \sigma \in \mathbb{C}$, the same branch convention gives

$$\begin{aligned} e^{\rho z+\sigma\zeta} &= e^{(\rho+\sigma)u+i\rho v}\exp\left(\frac{i\sigma}{\lambda}\mathrm{Log}_\mu W(a)\right)\\ &= e^{(\rho+\sigma)u+i\rho v}W(a)^{i\sigma/\lambda}. \end{aligned} \tag{159}$$

This is a pointwise CR family. The calculation supplies no Hilbert-space orthogonality or completeness statement.

B.6 Filtered affine–Appell calculations

Let $r = e^{\lambda v}$ and $B_n = r^{-n}a^n$. Since $D_v r = \lambda r$, $D_v a = \lambda a$, and $D_u a = \mu$, one finds

$$D_u B_n = \mu n r^{-1} B_{n-1}, \tag{160}$$

$$D_v B_n = 0, \tag{161}$$

$$T B_n = n r^{-1} B_{n-1}. \tag{162}$$

If $P \in C^\infty(\mathbb{R}^2_{(u,v)}, \mathbb{C})$, Leibniz' rule therefore gives the module actions in Proposition 4.16. The factor r^{-1} is a smooth coefficient, but it is not a complex scalar; this is exactly why the correct structure is a filtered $C^\infty(\mathbb{R}^2_{(u,v)}, \mathbb{C})$-module rather than a finite-dimensional basis.

There is nevertheless a finite algebraic antiderivative for coefficients of finite u-degree. Write $(n+1)_0 = 1$ and $(n+1)_k = (n+1)(n+2)\cdots(n+k)$ for $k \geq 1$.

Proposition B.1 (Finite upward sweep)**.** *Let $P \in C^\infty(\mathbb{R}^2_{(u,v)}, \mathbb{C})$ satisfy $\partial_u^{m+1}P = 0$ for some $m \geq 0$. For $n \geq 0$, define*

$$\mathcal{I}_u(PB_n) = \sum_{k=0}^{m}(-1)^k \frac{r^{k+1}\partial_u^k P}{\mu^{k+1}(n+1)_{k+1}} B_{n+k+1}. \tag{163}$$

Then $\mathcal{I}_u(PB_n)$ belongs to $\mathcal{M}_{n+m+1}$ and

$$D_u\mathcal{I}_u(PB_n) = PB_n. \tag{164}$$

Proof. Denote the kth summand in Equation (163) by S_k. Because $D_u r = 0$, Leibniz' rule and Equation (160) give

$$\begin{aligned} D_u S_k = (-1)^k \frac{r^{k+1}\partial_u^{k+1} P}{\mu^{k+1}(n+1)_{k+1}} B_{n+k+1} \\ + (-1)^k \frac{r^k \partial_u^k P}{\mu^k (n+1)_k} B_{n+k}. \end{aligned}$$

The second term for $k = 0$ is PB_n. For $0 \le k < m$, the first term for k cancels the second term for $k+1$. The only remaining first term is the one with $k = m$, and it vanishes because $\partial_u^{m+1} P = 0$. This proves Equation (164). □

The operator $\mathcal{I}_u$ in this proposition is only a finite algebraic right inverse on the stated polynomial-in-u coefficient class. It is not a bounded inverse on either contact L^2 space and gives no density, basis, or completeness theorem for the filtered module.

C Poisson and Fourier calculations

This appendix collects the scale and Fourier checks underlying Sections 6–8. They are included to make the boundary normalization independently auditable.

C.1 Metric, volume, and Laplacian under the scaled coordinate

Set $X = (\lambda/\mu)x$. Then

$$dx = \frac{\mu}{\lambda} dX, \qquad \frac{dx^2}{\mu^2} = \frac{dX^2}{\lambda^2},$$

and hence

$$g_{\mu,\lambda} = \frac{1}{\lambda^2 y^2}(dX^2 + dy^2).$$

The inverse matrix and volume density in (x, y) are

$$g^{xx} = \mu^2 y^2, \qquad g^{yy} = \lambda^2 y^2, \qquad \sqrt{\det g} = \frac{1}{\mu\lambda y^2}.$$

Since

$$\sqrt{\det g}\, g^{xx} = \frac{\mu}{\lambda}, \qquad \sqrt{\det g}\, g^{yy} = \frac{\lambda}{\mu},$$

the divergence-of-gradient formula gives

$$\Delta_g = \mu^2 y^2 \partial_x^2 + \lambda^2 y^2 \partial_y^2 = \lambda^2 y^2 (\partial_X^2 + \partial_y^2).$$

Similarly,

$$\begin{aligned} |dF|_g^2 \mathrm{dvol}_g &= \left(\mu^2 y^2 |F_x|^2 + \lambda^2 y^2 |F_y|^2\right) \frac{dx\, dy}{\mu\lambda y^2} \\ &= \left(\frac{\mu}{\lambda}|F_x|^2 + \frac{\lambda}{\mu}|F_y|^2\right) dx\, dy \\ &= (|F_X|^2 + |F_y|^2) dX\, dy. \end{aligned}$$

C.2 Transport of the Poisson density

Starting from

$$P_y(X-t)dt = \frac{1}{\pi}\frac{y}{(X-t)^2+y^2}\,dt,$$

substitute

$$X = \frac{\lambda}{\mu}x, \qquad t = \frac{\lambda}{\mu}\xi, \qquad dt = \frac{\lambda}{\mu}d\xi.$$

Then

$$\begin{aligned} P_y(X-t)dt &= \frac{1}{\pi}\frac{y(\lambda/\mu)}{(\lambda^2/\mu^2)(x-\xi)^2+y^2}\,d\xi \\ &= \frac{\mu\lambda y}{\pi\big(\lambda^2(x-\xi)^2+\mu^2y^2\big)}\,d\xi. \end{aligned}$$

This proves both the original-coordinate formula and its normalization, because the change of boundary coordinate includes the density factor dt.

C.3 Fourier transform of the kernel

For $h>0$, direct inversion gives

$$\begin{aligned} \frac{1}{2\pi}\int_{\mathbb{R}} e^{i\kappa s}e^{-h|\kappa|}\,d\kappa &= \frac{1}{\pi}\operatorname{Re}\int_0^\infty e^{-(h-is)\kappa}\,d\kappa \\ &= \frac{1}{\pi}\operatorname{Re}\frac{1}{h-is} = \frac{1}{\pi}\frac{h}{s^2+h^2}. \end{aligned}$$

Equivalently,

$$\int_{\mathbb{R}} e^{-i\kappa s}\frac{1}{\pi}\frac{h}{s^2+h^2}\,ds = e^{-h|\kappa|}. \tag{165}$$

Taking $h=(\mu/\lambda)y$ shows that

$$\widehat{P_y^{\mu,\lambda}}(\kappa) = e^{-(\mu/\lambda)|\kappa|y}.$$

Thus, for Schwartz data,

$$\widehat{U}(\kappa,y) = \widehat{\varphi}(\kappa)e^{-(\mu/\lambda)|\kappa|y}.$$

Each mode satisfies

$$\begin{aligned} \left(\mu^2y^2\partial_x^2+\lambda^2y^2\partial_y^2\right)e^{i\kappa x-(\mu/\lambda)|\kappa|y} &= y^2\left(-\mu^2\kappa^2+\lambda^2\frac{\mu^2}{\lambda^2}\kappa^2\right)e^{i\kappa x-(\mu/\lambda)|\kappa|y} \\ &= 0, \end{aligned}$$

which is a direct Fourier verification of harmonicity.

C.4 Energy and conormal scale

Put $c=\mu/\lambda$. Plancherel and the preceding transform give

$$\begin{aligned} \mathcal{E}_g(U) &= \frac{1}{2\pi}\int_{\mathbb{R}}\int_0^\infty \left(c|i\kappa\widehat{U}|^2+c^{-1}|\partial_y\widehat{U}|^2\right)dy\,d\kappa \\ &= \frac{1}{2\pi}\int_{\mathbb{R}}\int_0^\infty 2c\kappa^2|\widehat{\varphi}(\kappa)|^2e^{-2c|\kappa|y}\,dy\,d\kappa \\ &= \frac{1}{2\pi}\int_{\mathbb{R}}|\kappa|\,|\widehat{\varphi}(\kappa)|^2\,d\kappa. \end{aligned}$$

The raw vertical derivative is

$$-U_y(\,\cdot\,,0) = c|D_x|\varphi.$$

On $y = \varepsilon$, however,

$$\nu = -\lambda y \partial_y, \qquad d\sigma_g = \frac{dx}{\mu y},$$

so the boundary term in Green's identity is

$$(\partial_\nu U) d\sigma_g = -\frac{\lambda}{\mu} U_y \, dx.$$

Consequently the variational conormal derivative is

$$\mathcal{N}_x \varphi = -\frac{\lambda}{\mu} U_y(\,\cdot\,,0) = |D_x|\varphi,$$

with no remaining μ or λ factor. In the normalized X-coordinate, the same cancellation appears as the familiar identity $-U_y(\cdot, 0) = |D_X|\psi$.

C.5 Assignment-only fields and the logarithmic example

For completeness, the direct chain-rule calculation is

$$\Delta_g(h(a)) = h''(a)|\nabla a|_g^2 + h'(a)\Delta_g a = (\mu^2 + \lambda^2 a^2) h''(a) + 2\lambda^2 a h'(a).$$

Writing $p = h'$ reduces the harmonic equation to

$$\frac{p'}{p} = -\frac{2\lambda^2 a}{\mu^2 + \lambda^2 a^2},$$

whence

$$p(a) = \frac{C}{\mu^2 + \lambda^2 a^2}, \qquad h(a) = C_0 + C_1 \arctan\left(\frac{\lambda a}{\mu}\right).$$

Finally, since

$$w = y\left(i - \frac{\lambda}{\mu} a\right),$$

the principal logarithm on $\mathbb{H}$ is

$$\operatorname{Log} w = \log y + \frac{1}{2}\log\left(1 + \frac{\lambda^2 a^2}{\mu^2}\right) + i\left(\frac{\pi}{2} + \arctan\frac{\lambda a}{\mu}\right).$$

Its imaginary part divided by π is the negative-half-line harmonic measure. The polar-coordinate energy calculation

$$\int_0^\pi \int_0^\infty r^{-2} r \, dr \, d\theta = \infty$$

confirms directly that this bounded solution is not in the global finite-energy class used for the Dirichlet-to-Neumann theorem.

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